

GIFT GALLERY: Gift and Accessories Store

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

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PROJECT

DEPARTMENT OF COMPUTER SCIENCE

SESSION: 2023 - 2025 | GOVT.POST GRADUATE KHAWAJA FAREED COLLEGE, RAHIM YAR KHAN

Revision History

Date	Description	Author	Comments
	Versions 1	LAIBA YASEEN	

Document Approval

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1. Introduction

1.1 Purpose

The Software Requirements Specification (SRS) document serves as a comprehensive blueprint for developing a software system. Its primary purpose is to define the software's functional and non-functional requirements, providing a clear and detailed description of what the system should accomplish. The SRS facilitates effective communication among stakeholders, including developers, project managers, and clients, by outlining the expectations and constraints. It acts as a reference throughout the development process, ensuring that the final product aligns with the initial vision and requirements.

1.2 Scope

This project includes developing an intuitive e-commerce website: featuring diverse product categories, customization options, secure shopping, multiple payment options, and fast delivery. It will offer user accounts, advanced search and filter functions, a streamlined checkout process, and integration with shipping providers. The site will also include an admin dashboard for product management and marketing tools to enhance visibility and sales. Mobile responsiveness and SEO optimization will ensure broad accessibility and engagement.

1.3 Definitions, Acronyms, and Abbreviations.

GAS =Gift and Accessories Store.

SRS = Software Requirements Specification.

GUI = Graphical User Interface

UML = Uniformed Modifying Language

FAQ = Frequently Asked Questions

1.4 References

- IEEE SRS Format.

1.5 Overview

This section describes the remaining section of SRS. Section 2 of the SRS describes the product in more detail. Section 3 provides a complete list of the function requirements of the system. Section 4 provides the non-functional requirements. Section 5 show the class diagram, and Section 6 the use case diagram.

2. The Overall Description

2.1 Product Perspective

GiftGallery will serve as a premier online marketplace for unique and customizable gifts, offering a seamless and engaging shopping experience. The platform will integrate secure payment gateways, efficient inventory management, and advanced customization options. Designed for scalability and mobile responsiveness, GiftGallery ensures a smooth user experience across all devices while maintaining robust security measures to protect transactions and user data.

The following subsections describe how the software operates inside various constraints.

2.1.1 Operations

In a **Software Requirements Specification (SRS)** for **GiftGallery**, the operations from a product perspective include:

1. Product Management

- Add, update, delete products.
- Categorize products and manage inventory.

2. User Operations

- Browse, search, filter products.
- Add items to cart/wishlist, personalize gifts, and complete purchases.

3. Order Management

- View order status, manage cancellations/returns, track deliveries.

4. Discounts and Promotions

- Manage coupons, gift cards, and discounts.

5. Review and Rating System

- Allow product reviews and ratings.

6. Notifications

- Send stock alerts, order updates, and delivery notifications.

7. Reporting and Analytics

- Generate sales, inventory, and customer behavior reports.

8. Integrations

- Payment gateways and shipping services integration.

These operations ensure a smooth flow of product management, user interaction, and business processes in GiftGallery.

2.1.2 Site Adaptation Requirements

In the **SRS for GiftGallery**, **Site Adaptation Requirements** from a product perspective include:

1. **Localization:** Support multiple languages, currencies, and region-specific pricing/taxes.
2. **Responsive UI:** Adapt UI for different devices and cultural preferences.
3. **Payment Integration:** Use region-specific payment gateways.
4. **Shipping:** Customize shipping options and rates by location.
5. **Legal Compliance:** Adhere to local privacy and tax laws.

These ensure the platform adapts to different markets and user needs.

2.2 Product Functions

GiftGallery will provide a suite of essential functions to deliver a top-notch user experience. Users will enjoy intuitive product browsing and search features, allowing them to easily find and filter gifts by various criteria. The platform will offer robust customization options for personalizing items, with real-time previews before purchase. A secure and streamlined checkout process will support multiple payment methods, ensuring smooth transactions. Users can manage their accounts, track orders, and view order histories. The admin dashboard will facilitate efficient product and order management, while marketing tools will enable the creation of promotional campaigns and discount

codes. Additionally, customer reviews and ratings will help enhance product credibility and assist other shoppers in making informed decision

2.3 User Characteristics

In GiftGallery's SRS, the **User Characteristics** include:

1. **Customers:** General shoppers needing easy navigation, search, and secure checkout.
2. **Admins:** Platform managers handling inventory, orders, and analytics.
3. **Sellers (Optional):** Merchants managing product listings, sales, and payments.
4. **Guests:** Unregistered users browsing with limited access.

These guide the system design for different user needs.

2.4 General Constraints

In GiftGallery's SRS, the General Constraints may include:

Budget Limitations: Restricted resources for development, hosting, and maintenance.

1. **Time Constraints:** Deadlines for launch and feature updates.
2. **Compliance:** Adherence to privacy (GDPR) and tax regulations.
3. **Performance:** Must handle high traffic and transactions efficiently.
4. **Compatibility:** Support for various devices, browsers, and payment gateways.
5. **Security:** Ensure data protection and secure transactions.

These constraints influence the system's design and development process.

2.5 Assumptions and Dependencies

No specific software, hardware and network assumptions and dependencies are required.

All required technologies are available..

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 System Interfaces

The **System Interfaces** detail interactions with external components, including payment gateways like PayPal and Stripe for secure transactions, shipping providers for order fulfillment,

and third-party marketing tools for customer engagement. These interfaces enable seamless communication and enhance user experience and operational efficiency.

3.1.2 Interfaces

The **Interfaces** in the external interface requirements outline the connections between the system and external entities. This includes user interfaces for customers and admins, APIs for integrating payment gateways (e.g., PayPal, Stripe) and shipping providers, and interactions with external databases and analytics tools. These interfaces enable smooth operation and effective communication with external systems.

3.1.3 Hardware Interfaces

The **Hardware Interfaces** in the external interface requirements specify the physical and network components required for system operation. This includes servers for hosting the platform, which must support web applications and database management. Additionally, it encompasses network infrastructure such as routers and firewalls to ensure secure and reliable connectivity for users accessing the platform. The hardware interfaces ensure the system runs efficiently and can handle expected traffic loads while maintaining performance and security.

3.1.4 Software Interfaces

The **Software Interfaces** in the external interface requirements detail the interactions between the system and various software components. This includes integration with third-party APIs for payment processing (e.g., PayPal, Stripe) and shipping services, as well as connections to external databases for user data management. The system will also interface with content management systems (CMS) for managing product listings and marketing tools for customer engagement. These software interfaces ensure seamless communication and functionality across the GiftGallery platform and its external services.

3.1.5 Communications Interfaces

The **Communication Interfaces** in the external interface requirements outline the protocols and methods used for data exchange between the system and external entities. This includes internet protocols such as HTTP/HTTPS for secure web communications, as well as RESTful APIs for

integrating with payment gateways and shipping services. The communication interfaces also encompass email protocols (SMTP) for sending notifications and confirmations to users. These interfaces ensure reliable and secure data transmission, facilitating effective communication within the platform and with external services.

3.2 Functional Requirements

The **Functional Requirements** of **GiftGallery** in the SRS documentation outline the specific behaviors and functionalities the system must exhibit.

3.2.1 User Registration and Authentication

3.2.1.1 *Introduction:*

Users must be able to create an account and log in to access the personalized features of GiftGallery.

3.2.1.2 *Inputs:*

User's email address

Password

Optional: Name, phone number

3.2.1.3 *Processing:*

Validate email format.

Check for existing accounts.

Hash the password for security.

Store user details in the database.

3.2.1.4 *Outputs:*

Success message upon successful registration or login.

Error messages for failed login (e.g., incorrect password) or registration (e.g., email already in use).

3.2.1.5 Error Handling:

If the email format is invalid, prompt the user to enter a valid email.

Display a message for incorrect login attempts or duplicate registration.

3.2.2 Product Search and Filtering

3.2.2.1 Introduction:

Users should be able to search for products and apply filters to find desired items efficiently.

3.2.2.2 Inputs:

Search keywords

Selected filters (e.g., category, price range, ratings)

3.2.2.3 Processing:

Query the database using the provided keywords and filters.

Sort results based on user preferences (e.g., relevance, price).

3.2.2.4 Outputs:

A list of products matching the search criteria.

Pagination controls for navigating through search results.

3.2.2.5 **Error Handling:**

If no products are found, display a message suggesting alternative searches or filters.

Handle database connection errors and display a user-friendly error message.

3.2.3 **Shopping Cart Management**

3.2.3.1 **Introduction:**

Users must be able to complete their purchases through a secure checkout process.

3.2.3.2 **Inputs:**

Shipping address

Payment information (credit card details, billing address)

3.2.3.3 **Processing:**

Validate input data for correctness and completeness.

Process payment through a secure gateway.

Create an order record in the database.

3.2.3.4 **Outputs:**

Order confirmation with a unique order ID.

Email receipt sent to the user.

3.2.3.5 **Error Handling:**

If payment fails, notify the user and prompt for a different payment method.

Display validation errors for incorrect shipping or payment information.

3.2.4 **Checkout Process**

3.2.4.1 **Introduction:**

Users must be able to complete their purchases through a secure checkout process.

3.2.4.2 Inputs:

Shipping address

information (credit card details, billing address)

3.2.4.3 Processing:

Validate input data for correctness and completeness.

Process payment through a secure gateway.

Create an order record in the database.

3.2.4.4 Outputs:

Order confirmation with a unique order ID.

Email receipt sent to the user.

3.2.4.5 Error Handling:

If payment fails, notify the user and prompt for a different payment method.

Display validation errors for incorrect shipping or payment information.

3.2.5 Review and Rating Submission

3.2.5.1 Introduction:

Users can submit reviews and ratings for products they have purchased.

3.2.5.2 Inputs:

Product ID

Rating (e.g., 1 to 5 stars)

Review text

3.2.5.3 Processing:

Validate the product ID and check if the user has purchased the product.

Store the review and rating in the database associated with the product.

3.2.5.4 **Outputs:**

Success message confirming review submission.

Display the updated average rating and new review on the product page.

3.2.5.5 **Error Handling:**

If the user hasn't purchased the product, inform them they cannot leave a review.

Handle database errors during submission and provide an error message

3.2.6 **Order Tracking**

3.2.6.1 **Introduction:**

Users should be able to track the status of their orders post-purchase.

3.2.6.2 **Inputs:**

Order ID

3.2.6.3 **Processing:**

Retrieve order details from the database using the order ID.

Check the current status of the order (e.g., processing, shipped, delivered).

3.2.6.4 **Outputs:**

Display order status, tracking information, and estimated delivery date.

3.2.6.5 **Error Handling:**

If the order ID is invalid, display a message indicating no such order exists.

Handle any database connection errors gracefully.

3.2.7 **Wishlist Functionality**

3.2.7.1 **Introduction:**

Users can create and manage a wishlist for products they are interested in purchasing later.

3.2.7.2 **Inputs:**

Product ID

Action (add or remove from wishlist)

3.2.7.3 **Processing:**

Validate the product ID.

Update the user's wishlist in the database.

3.2.7.4 **Outputs:**

Updated wishlist displaying current items.

Notifications confirming additions or removals.

3.2.7.5 **Error Handling:**

If the product ID is invalid, inform the user with an error message.

Handle database errors while updating the wishlist gracefully.

3.2.8 **User Profile Management**

3.2.8.1 **Introduction:**

Users should be able to view and update their profile information, including personal details and payment methods.

3.2.8.2 **Inputs:**

User information (name, email, phone number, address)

Payment details (credit card information)

3.2.8.3 **Processing:**

Validate the updated information for correctness.

Update the user profile in the database.

3.2.8.4 **Outputs:**

Success message confirming the updates.

Display the updated profile information.

3.2.8.5 Error Handling:

Provide an error message if the input data is invalid (e.g., incorrect email format).

Handle any database update errors and inform the user appropriately

3.2.9 Promotions and Discount Management

3.2.9.1 Introduction:

The system must support creating and managing promotional offers and discount codes.

3.2.9.2 Inputs:

Discount code

Promotion details (percentage, start/end dates)

3.2.9.3 Processing:

Validate the discount code against existing promotions.

Apply the discount to the user's cart during checkout.

3.2.9.4 Outputs:

Confirmation of discount application and updated total price.

Error message if the discount code is invalid or expired.

3.2.9.5 Error Handling:

Inform the user if the code is not applicable or has expired.

Handle any issues with the promotion database gracefully.

3.2.10 Notifications and Alerts

3.2.10.1 Introduction:

Users should receive notifications for important events, such as order confirmations, shipping updates, and promotional offers.

3.2.10.2 Inputs:

User preferences for notification settings (email, SMS)

3.2.10.3 Processing:

Trigger notifications based on user actions and system events (e.g., order placed, order shipped).

Store notification preferences in the user profile.

3.2.10.4 Outputs:

Emails or SMS alerts based on the triggered events.

Notification logs for user reference.

3.2.10.5 Error Handling:

If a notification fails to send, log the error and attempt to resend.

Inform users of any issues with their preferred notification method.

3.2.11 Analytics and Reporting

3.2.11.1 Introduction:

Admins should be able to generate reports on various metrics to analyze business performance.

3.2.11.2 Inputs:

Report type (sales, inventory, user activity)

Date range for the report

3.2.11.3 Processing:

Query the database for the selected report type and date range.

Aggregate and summarize the data.

3.2.11.4 Outputs:

Generate and display reports in a readable format (e.g., charts, tables).

Option to export reports in different formats (CSV, PDF).

3.2.11.5 Error Handling:

If no data is available for the selected criteria, inform the admin.

Handle any database query errors and provide an appropriate error message.

3.2.12 Security Features

3.2.12.1 Introduction:

The system must implement security measures to protect user data and transactions.

3.2.12.2 Inputs:

User credentials (for authentication)

Payment information (during checkout)

3.2.12.3 Processing:

Encrypt sensitive data during storage and transmission.

Validate user permissions for sensitive actions (e.g., admin access).

3.2.12.4 Outputs:

Secure access to user accounts and administrative functions.

Confirmation of secure transactions.

3.2.12.5 Error Handling:

If authentication fails, provide a message indicating incorrect credentials.

Handle any security breaches by logging the incident and notifying admins.

3.2.13 Multi-language and Localization Support

3.2.13.1 Introduction:

The system should support multiple languages and localization settings based on user preferences.

3.2.13.2 Inputs:

User-selected language preference

3.2.13.3 Processing:

Adjust the user interface and content based on the selected language.

Localize date formats, currencies, and measurement units.

3.2.13.4 Outputs:

Display the platform in the selected language with appropriate local settings.

3.2.13.5 Error Handling:

If a language pack is not available, default to a primary language and notify the user.

Handle localization errors gracefully.

3.2.14 Feedback and Support System

3.2.14.1 Introduction:

Users should be able to submit feedback and access support for issues they encounter.

3.2.14.2 Inputs:

Feedback text

User contact information

3.2.14.3 Processing:

Store user feedback in the database for review.

Route support requests to the appropriate support team.

3.2.14.4 **Outputs:**

Confirmation message acknowledging receipt of feedback or support request.

Follow-up communication regarding the resolution of issues.

3.2.14.5 **Error Handling:**

If feedback submission fails, inform the user and allow for resubmission.

Log issues with support routing for internal review.

3.2.15 **Virtual Gift Wrapping with Customization**

3.2.15.1 **Introduction:**

Allows users to virtually wrap gifts by selecting wrapping designs, colors, and tags, with a preview option before finalizing.

3.2.15.2 **Inputs:**

Gift item (product ID)

Wrapping style, color, and tag message

3.2.15.3 **Processing:**

Validate wrapping options for the selected gift.

Generate a virtual 3D preview of the wrapped gift with user choices.

3.2.15.4 **Outputs:**

A customized digital preview of the wrapped gift.

Confirmation of wrapping selection in the order summary.

3.2.15.5 **Error Handling:**

If a wrapping style is unavailable, suggest alternatives.

Notify the user if the selected design is incompatible with the gift size.

3.2.16 Social Shopping with Live Interaction

3.2.16.1 Introduction:

Enables users to shop together with friends or family through live video or chat, allowing real-time discussions and collective decision-making.

3.2.16.2 Inputs:

User login details

Participants' contact (emails or phone numbers)

Product links shared in the session

3.2.16.3 Processing:

Create a shared virtual shopping session.

Synchronize product views and enable live interaction via video, chat, or reactions.

3.2.16.4 Output:

A collaborative shopping session link.

Finalized cart items chosen collectively.

3.2.16.5 Error Handling:

If a participant fails to join, provide troubleshooting steps or an alternative link.

If synchronization fails, allow users to refresh and rejoin the session.

3.2.17 Group Gifting with Split Payments

3.2.17.1 **Introduction:**

This feature allows multiple users to contribute to a single gift purchase. Users can invite friends or family to pool funds for a gift, splitting the cost at checkout.

3.2.17.2 **Inputs:**

Gift selection (product ID)

List of contributors (email or phone numbers)

Contribution amounts

3.2.17.3 **Processing:**

System calculates the total cost and divides it among contributors.

Generates a payment link for each contributor.

Tracks contributions until the total amount is met.

3.2.17.4 **Outputs:**

Notifications sent to contributors with payment links.

Confirmation message once the gift is fully funded.

3.2.17.5 **Error Handling:**

If a contributor fails to pay, notify the organizer to adjust contributions or replace the unpaid share.

Timeout option if contributions aren't completed within a set period, allowing cancellation or refund.

3.2.18 **Personalized Gift Messages with Hand Replica**

3.2.18.1 **Introduction:**

This feature allows users to upload a sample of their handwriting, which is used to generate a digital version of a personalized gift message.

3.2.18.2 **Inputs:**

Text message input

Handwriting sample (image upload)

3.2.18.3 **Processing:**

Converts the user-uploaded handwriting sample into a digital font.

Combines the text message with the generated font to create a realistic handwritten note.

3.2.18.4 **Outputs:**

Digital gift message included with the order, downloadable or printable.

Option for the note to be printed on a card and shipped with the gift.

3.2.18.5 **Error Handling:**

If the handwriting sample upload fails, prompt the user to retry or choose a default handwriting style.

Provide a preview of the note for user approval before finalizing.

3.2.19 **Dynamic Packaging Options**

3.2.19.1 **Introduction:**

Users can customize their packaging, selecting options such as eco-friendly materials, theme-based wrapping, or even branded reusable packaging.

3.2.19.2 **Inputs:**

Product selection (product ID)

Packaging preferences (e.g., material, theme)

3.2.19.3 Processing:

Validates packaging options against product size and availability.

Updates the total cost based on the selected packaging.

3.2.19.4 Outputs:

Visual preview of the selected packaging.

Confirmation of the packaging choice with the order summary.

3.2.19.5 Error Handling:

If a selected packaging option is unavailable, suggest alternatives.

Display an error message for incompatible packaging sizes and products.

3.2.20 Multi-Destination Checkout

3.2.20.1 Introduction:

Users can purchase multiple items in a single order and send them to different addresses, ideal for sending gifts to multiple recipients.

3.2.20.2 Inputs:

Products selected (product IDs)

Delivery addresses for each product

3.2.20.3 Processing:

Splits the cart into sub-orders based on destination.

Calculates shipping costs separately for each address.

3.2.20.4 Outputs:

Breakdown of sub-orders by address.

Notifications sent to recipients upon shipment.

3.2.20.5 Error Handling:

If a delivery address is invalid, prompt the user to correct it.

Notify the user of any delays or issues with specific sub-orders.

3.2.21 Customizable Product Bundles

3.2.21.1 Introduction:

Users can create their product bundles by selecting complementary items, such as pairing a gift box with matching accessories or add-ons.

3.2.21.2 Inputs:

A product selection (product ID)

Add-ons or accessories (product IDs)

3.2.21.3 Processing:

Validates compatibility of selected products.

Applies discounts or bundle pricing if applicable.

3.2.21.4 Outputs:

Visual preview of the customized bundle.

Confirmation of bundle pricing and included items.

3.2.21.5 Error Handling:

If a selected add-on is out of stock, suggest alternatives.

Notify the user if the bundle exceeds size or weight limits for shipping.

3.2.22 **Instant Digital Gift Registry**

3.2.22.1 **Introduction:**

Users can create a digital gift registry for events like birthdays, weddings, or baby showers, allowing guests to purchase items directly from the list.

3.2.22.2 **Inputs:**

Event details (name, date, description)

Products added to the registry (product IDs)

3.2.22.3 **Processing:**

Generates a shareable link or QR code for the registry.

Tracks purchases and updates item availability in real-time.

3.2.22.4 **Outputs:**

Confirmation of registry creation and shareable link.

Notifications sent to the user when items are purchased.

3.2.22.5 **Error Handling:**

If an item in the registry goes out of stock, notify the user and suggest replacements.

Handle multiple purchases of the same item by restricting further purchases once the item is fulfilled.

3.2.23 **Charity Integration with Purchases**

3.2.23.1 **Introduction:**

Users can donate a percentage of their purchase to a selected charity directly at checkout.

3.2.23.2 Inputs:

Charity selection

Donation percentage or amount

3.3.23.3 Processing:

Calculates the donation based on the total purchase amount.

Transfers the donation to the selected charity upon transaction completion.

3.2.23.4 Outputs:

Confirmation of the donation with details of the beneficiary.

Option to download a receipt for tax purposes.

3.2.23.5 Error Handling:

If the charity's payment system fails, notify the user and suggest retrying or selecting another charity.

Provide an option to skip the donation step if issues persist.

3.2.24 Occasion-Based Gift Bundle

3.2.24.1 Introduction:

Allow customers to create or select pre-made gift bundles tailored to specific occasions like birthdays, anniversaries, or festivals. These bundles are curated based on complementary products.

3.2.24.2 Inputs:

Occasion type (e.g., birthday, wedding)

Budget range

Preferred categories (e.g., accessories, decorations)

3.2.24.3 **Processing:**

Filter products by the occasion and user preferences.

Combine selected items into a bundle and calculate total cost.

Offer customization options (e.g., add/remove items, adjust quantities).

3.2.24.4 **Outputs:**

Display the curated bundle with pricing and customization options.

Provide an option to save or purchase the bundle.

3.2.24.5 **Error Handling:**

If no products match the criteria, show a message suggesting alternative bundles.

If the selected bundle exceeds the budget, recommend adjustments.

3.2.25 **Delayed Delivery Scheduler**

3.2.25.1 **Introduction:**

Enable users to schedule gift deliveries in advance for special occasions, ensuring timely arrival.

3.2.25.2 **Inputs:**

Selected delivery date

Recipient's address

3.2.25.3 **Processing:**

Validate the selected date (e.g., ensure it is within allowable range).

Store the delivery details and schedule in the database.

3.2.25.4 **Outputs:**

Confirmation of the scheduled delivery date.

Email notification to the user closer to the delivery date.

3.2.25.5 Error Handling:

If the selected date is invalid or unavailable, suggest alternative dates.

Notify users if delivery cannot be completed due to external issues (e.g., holidays).

3.2.26 Multi-Recipient Order Splitting

3.2.26.1 Introduction:

Allow users to send items in a single order to multiple recipients with separate shipping addresses and messages.

3.2.26.2 Inputs:

List of recipient details (name, address, personalized message)

Item allocation per recipient

3.2.26.3 Processing:

Split the cart items into separate shipments.

Calculate shipping costs for each recipient's address.

3.2.26.4 Outputs:

Display a breakdown of shipments, costs, and delivery times.

Confirmation message summarizing all shipments.

3.2.26.5 Error Handling:

Notify the user if an address is invalid and allow corrections.

If order splitting fails, consolidate the order for a single recipient as a fallback.

3.2.27 Event-Based Discounts

3.2.27.1 Introduction:

Offer dynamic discounts during live events or countdowns, creating a sense of urgency for purchases.

3.2.27.2 Inputs:

Event type and duration (e.g., 2-hour flash sale)

Discount code or auto-applied discount

3.2.27.3 Processing:

Apply discounts to eligible products during the specified timeframe.

Display a countdown timer for the ongoing event.

3.2.27.4 Outputs:

Real-time updates of discounted prices in the cart.

Notification to users about the event and remaining time.

3.2.27.5 Error Handling:

If discounts fail to apply, provide a manual option for the user to claim the offer.

Notify users if the event has ended or if the product is out of stock.

3.2.28 Return And Refund Management

3.2.28.1 Introduction:

The system enables customers to request product returns or refunds through an automated process while notifying the administrator for review and action.

3.2.28.2 Inputs:

Customer-provided return reason, product details, and order ID.

Proof of purchase (e.g., receipt or order confirmation).

Optional photos for defective or damaged items.

3.2.28.3 Processing:

Validate return/refund request against store policies (e.g., within return window, product condition).

Notify administrators for approval or rejection.

Update the status of the return/refund in the system.

3.2.28.4 Outputs:

Confirmation of return/refund request submission.

Updates on the request status (e.g., approved, rejected, processed).

Refund completion notification to the customer.

3.2.28.5 Error Handling:

Display error messages for invalid requests (e.g., missing information or ineligible items).

Notify customers if the return period has expired.

Log system errors during processing for administrator review.

3.2.29 Customer Support and FAQ's

3.2.29.1 Introduction:

Enables users to access help through FAQs, submit queries, or contact support for assistance.

3.2.29.2 Inputs:

User-provided queries or issues.

Search keywords for FAQs.

Contact details (e.g., name, email).

3.2.29.3 Processing:

Match queries to relevant FAQ entries.

Route unresolved issues to support agents.

Generate and track support tickets.

3.2.29.4 Outputs:

FAQ responses or guidance.

Support ticket confirmation and updates.

Chat or email responses from agents.

3.2.29.5 Error Handling:

Notify users of submission failures or system errors.

Log issues for administrative review and prompt resolution.

3.2.30 Product Reviews and Ratings

3.2.30.1 Introduction:

Enables customers to share feedback about purchased products through reviews and ratings, enhancing customer trust and decision-making.

3.2.30.2 Inputs:

Customer-provided review text.

Numeric rating (e.g., 1–5 stars).

Optional media uploads (images or videos).

3.2.30.3 Processing:

Validate review input for completeness and appropriateness.

Link review to verified purchases.

Store review and rating in the database associated with the product.

3.2.30.4 Outputs:

Display average product ratings and individual reviews on the product page.

Sort or filter reviews (e.g., by date, rating).

3.2.30.5 Error Handling:

Display error messages for missing or invalid input fields.

Handle upload errors for unsupported or oversized media files.

Show notification for system issues during submission.

3.3 Use Cases

This section contains use cases of the Gift and Accessories store system.

3.3.1 Use Case: User Registration and Authentication

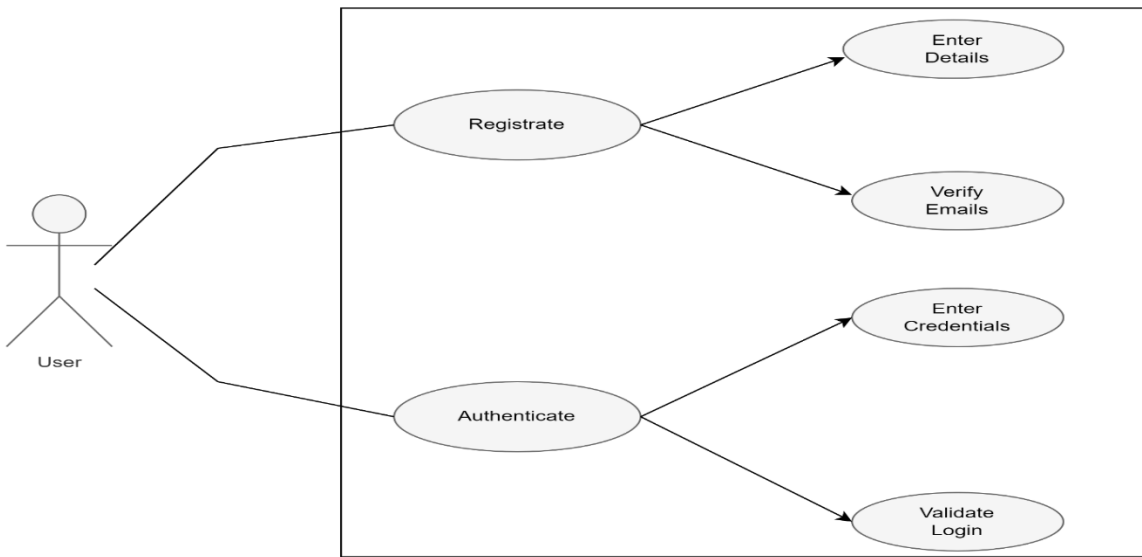


Figure 1 User Registration and Authentication Use Cas

Table 1 User Registration and Authentication Use Case

Use Case: User Registration

Actor	<i>User</i>
Pre-condition	<i>The user is not already registered in the system.</i>
Post-condition	<i>The user's account is created and ready for authentication.</i>
Steps	<i>1. User initiates the registration process.</i> <i>2. User enters personal details (e.g., name, email, password).</i> <i>3. System temporarily stores the user details.</i> <i>4. System sends a verification email to the user.</i>

3.3.2 Use Case: Product Search and Filtering

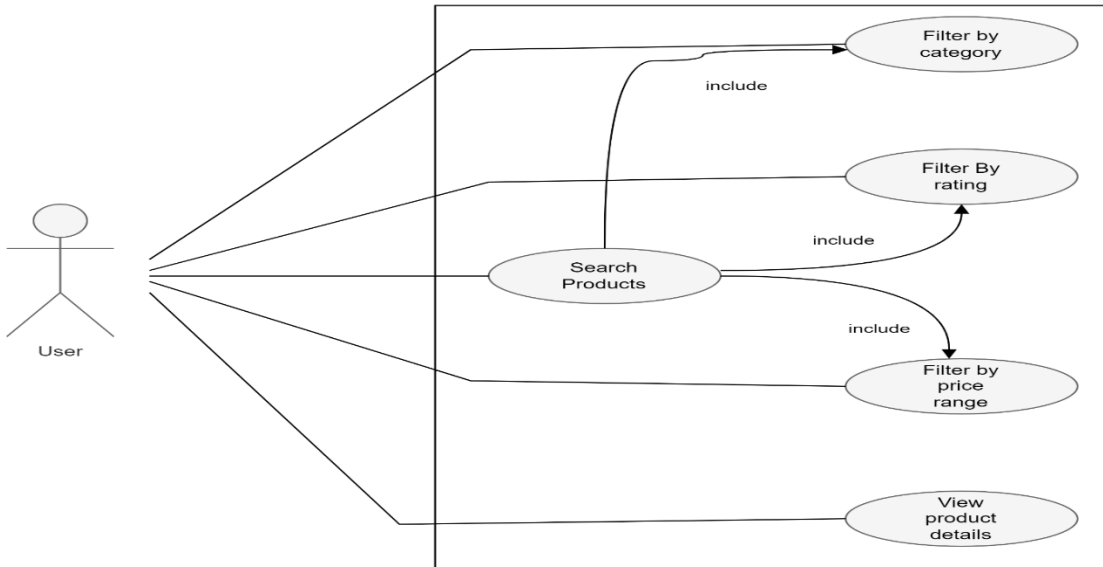


Figure 2 Product Search and Filtering Use Case

Table 2 Product Search and Filtering

Actor	<i>User</i>
Pre-condition	<i>User is on the product catalog or search page. Search results or product list is displayed.</i>
Post-condition	<i>Relevant product results are displayed. Filtered product list is shown to the user.</i>
Steps	1. User enters search keywords. 2. System retrieves matching products. 3. Results are displayed.

3.3.3 Use Case: Shopping Cart Management

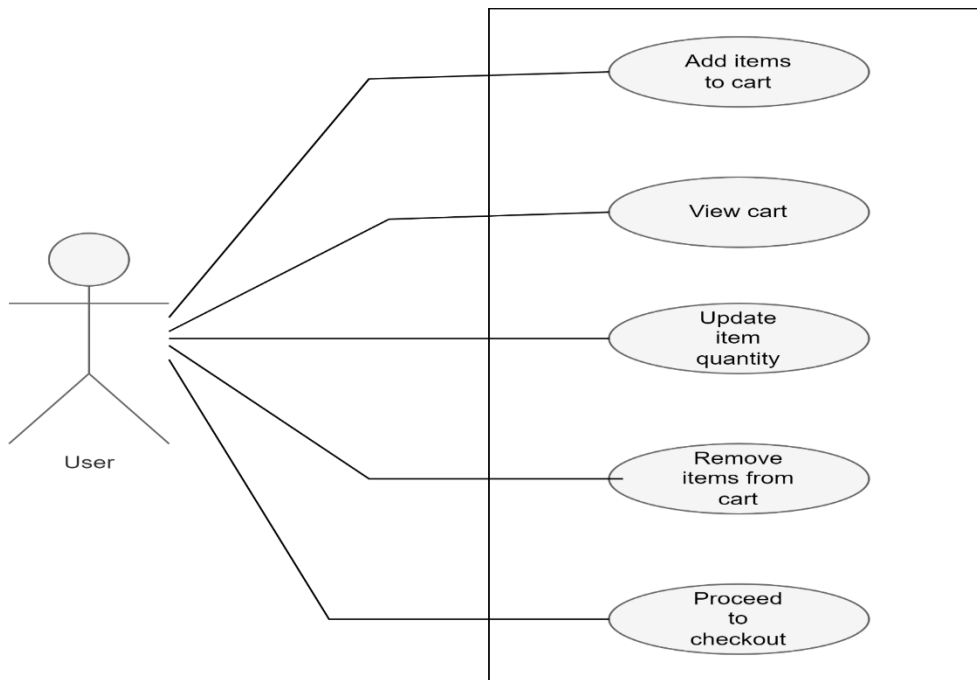


Figure 3 Shopping Cart Management Use case

Table 3 Shopping Cart Management

Actor	<i>User</i>
Pre-condition	<i>User is logged in and viewing a product. User has items in the cart.</i>
Post-condition	<i>Product is added to the cart. Cart contents are displayed. Checkout process is initiated.</i>
Steps	<i>1.User selects a product. 2.Cart contents are listed 3.User selects an item to remove. 4. Cart is updated with the new quantity.</i>

3.3.4 Use Case: Checkout Process

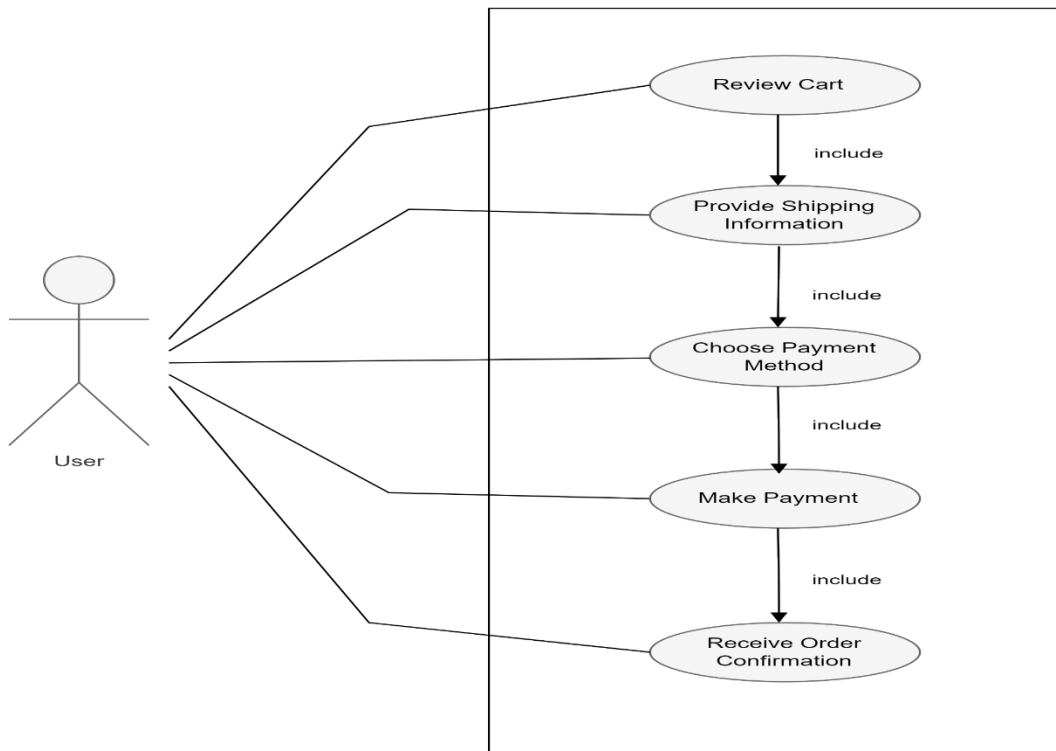


Figure 4 Checkout Process Use Case

Table 4 Checkout Process

Actor	<i>User</i>
Pre-condition	<i>User has items in the cart and is ready to purchase.</i>
Post-condition	<i>Order is placed successfully.</i>
Steps	<i>1. User reviews cart.</i> <i>2. User provides shipping information.</i> <i>3. User selects payment method.</i> <i>4. System processes the payment.</i> <i>5. Confirmation of order is displayed.</i>

3.3.5 Use Case: Review and Rating Submission

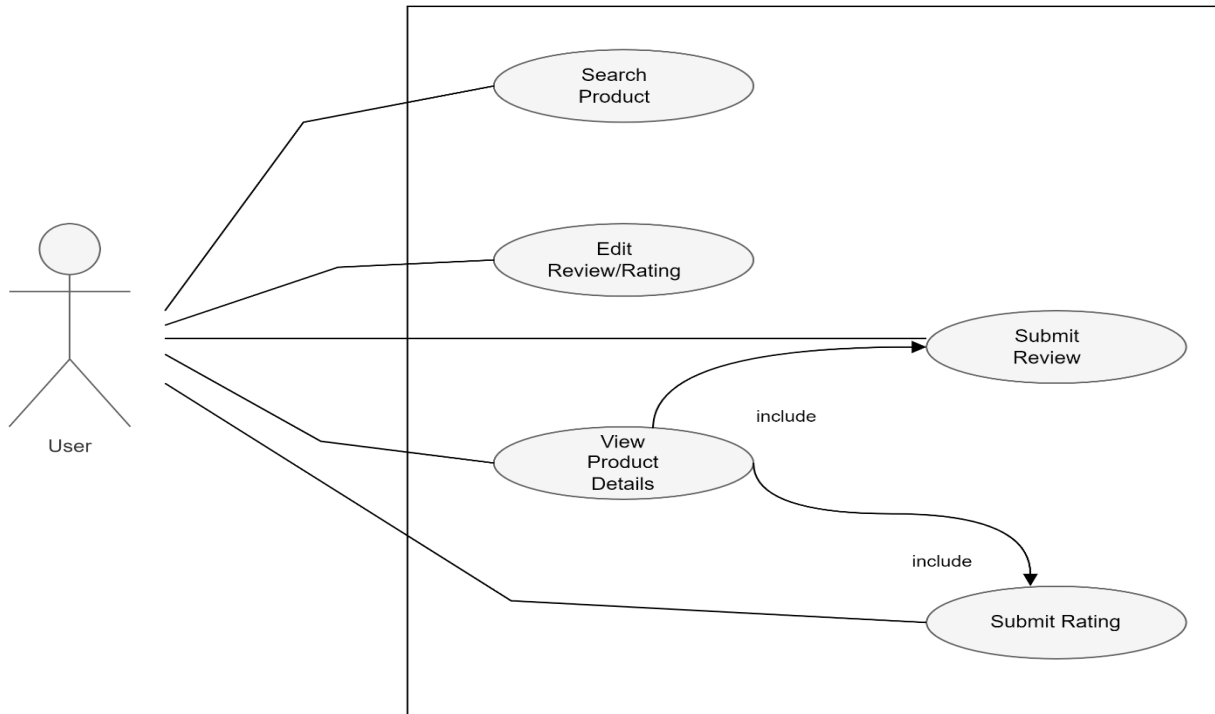


Figure 5 Review and Rating Submission Use Case

Table 5 Review and Rating Submission

Actor	<i>User</i>
Pre-condition	<i>User has purchased a product and is logged in.</i>
Post-condition	<i>Review and rating are submitted and saved in the system.</i>
Steps	<i>1. User navigates to the purchased product.</i> <i>2. User clicks on "Write a Review".</i> <i>3. User enters a review and selects a rating.</i> <i>4. User submits the review.</i> <i>5. System saves and displays the review.</i>

3.3.6 Use Case: Order Tracking

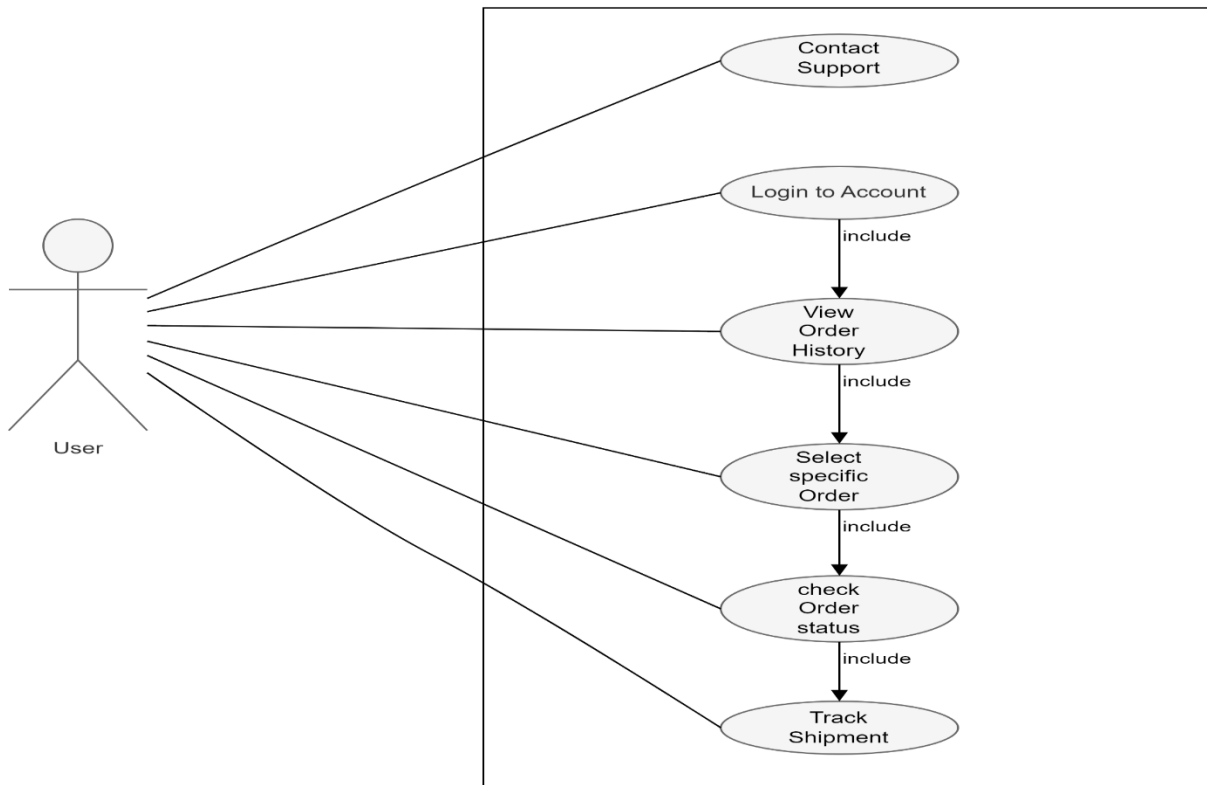


Figure 6 Order Tracking Use case

Table 6 Order Tracking

Actor	<i>User</i>
Pre-condition	<i>User has placed an order and is logged in.</i>
Post-condition	<i>Order status is displayed to the user.</i>
Steps	<i>1. User navigates to the "My Orders" section.</i> <i>2. User selects the specific order.</i> <i>3. System retrieves the order status.</i> <i>4. System displays the order status and tracking details.</i>

3.3.7 Use Case: Wishlist Functionality

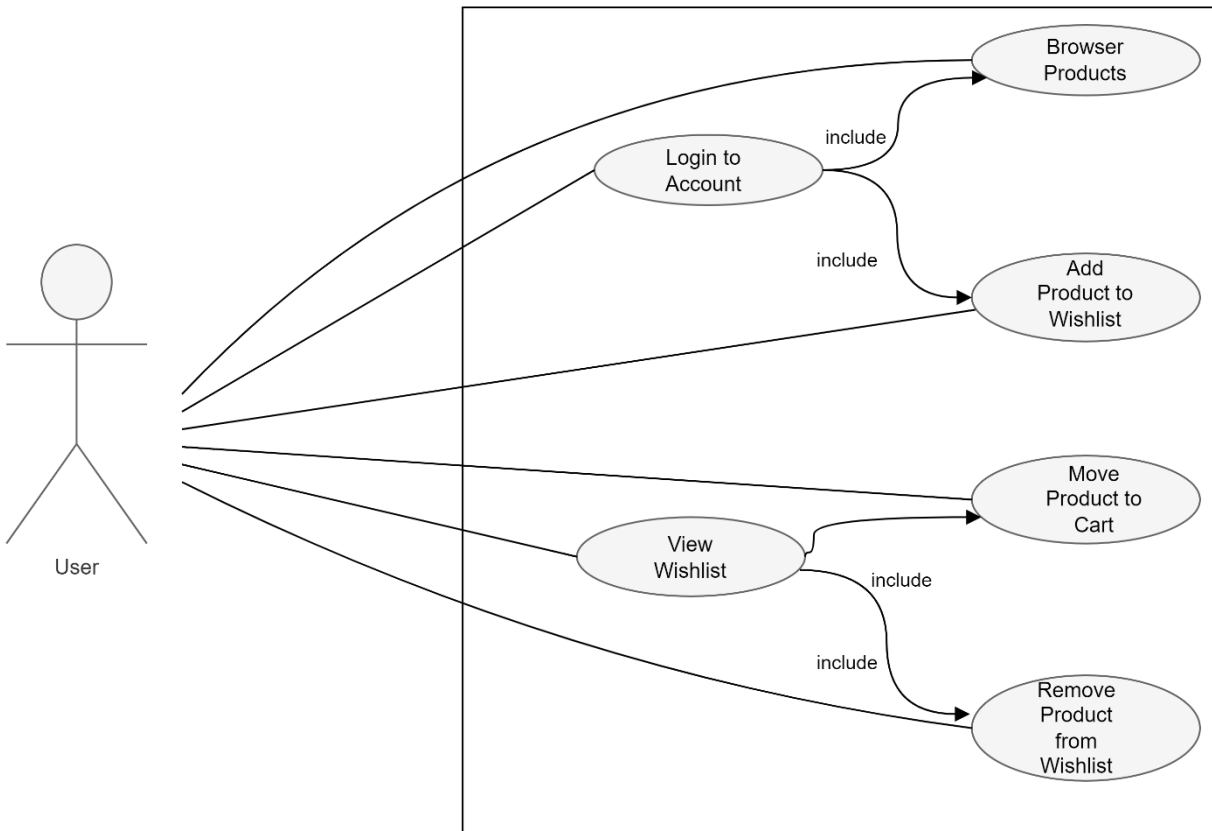


Figure 7 Wishlist Functionality Use Case

Table 7 Wishlist Functionality

Actor	<i>User</i>
Pre-condition	<i>User is logged in and viewing a product.</i>
Post-condition	<i>Product is added to the wishlist.</i>
Steps	1. User selects a product. 2. User clicks "Add to Wishlist". 3. System saves the product in the wishlist.

3.3.8 Use Case: User Profile Management

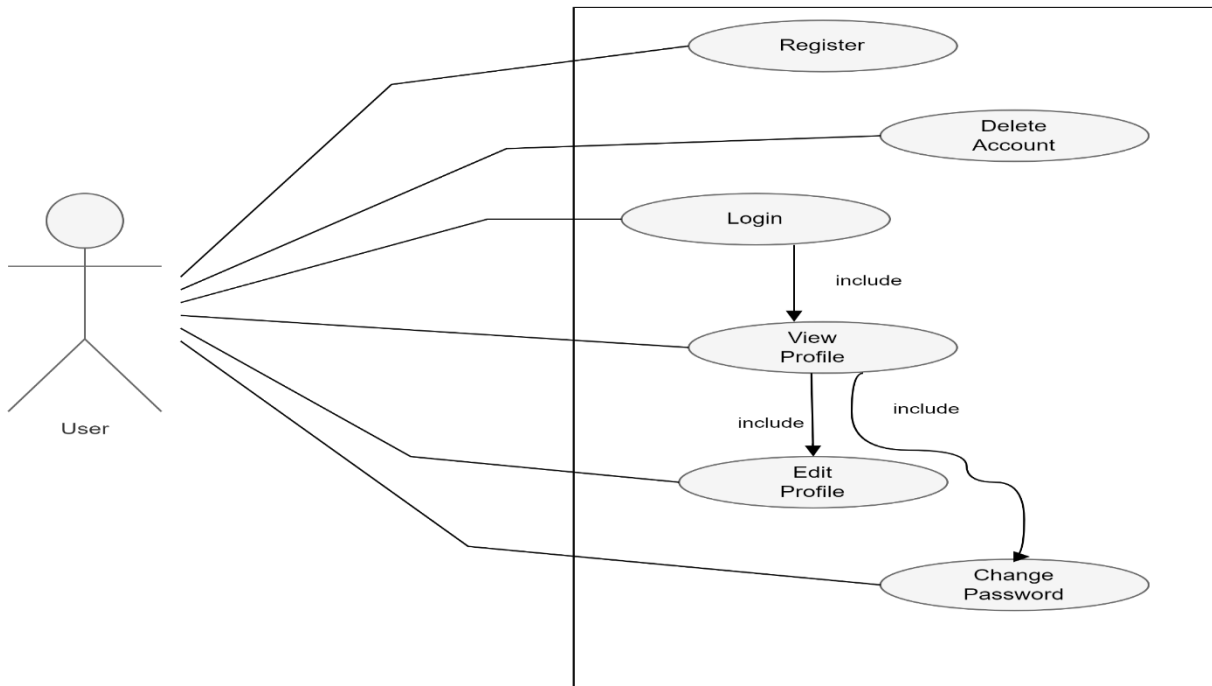


Figure 8 User Profile Management Use Case

Table 8 User Profile Management

Actor	User
Pre-condition	User is logged in. User is logged in and viewing their profile.
Post-condition	Profile details are displayed. Updated profile details are saved.
Steps	1. User navigates to the "Profile" section. 2. System retrieves and displays user details. 3. User clicks "Edit Profile". 4. User updates details (e.g., name, email). 5. System saves the changes. 6. User clicks "Delete Account".

3.3.9 Use Case: Promotions and Discounts

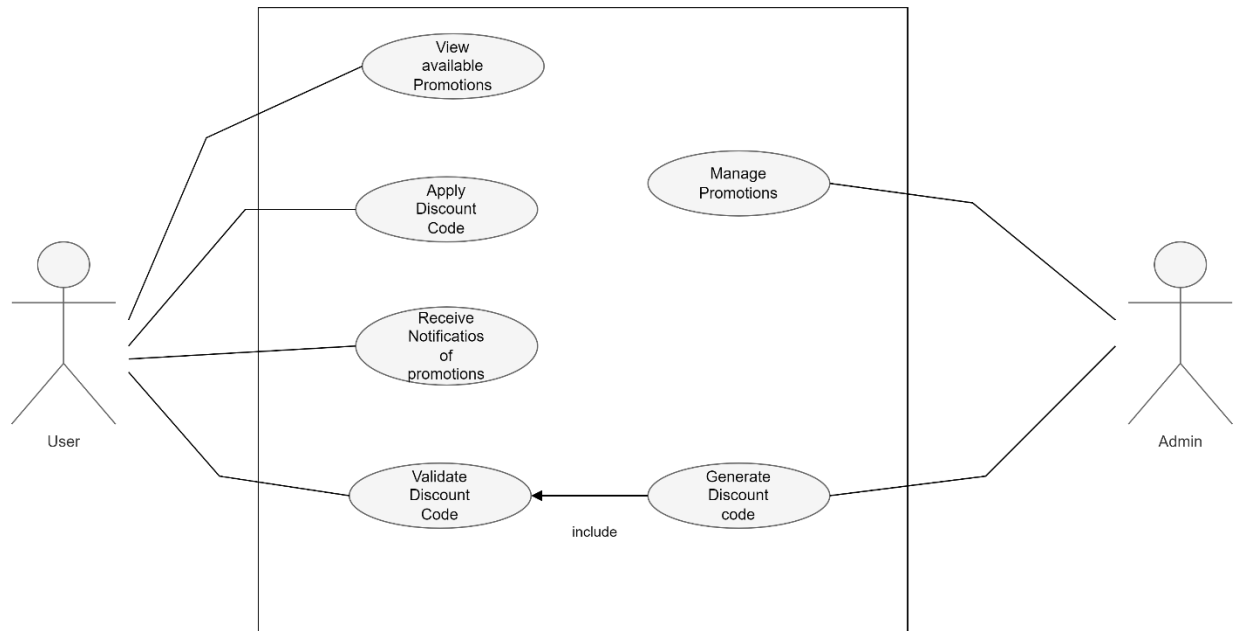


Figure 9 Promotions and Discounts Use case

Table 9 Promotions and Discounts

Actor	Admin/ User
Pre-condition	User is logged in and browsing the store. User has an active discount code and is on the checkout page.
Post-condition	Active promotions and discounts are displayed. Discount is applied to the cart total.
Steps	1. User navigates to the "Promotions" section. 2. System retrieves and displays active offers. 3. User enters the discount code. 4. System validates the code. 5. Discount is applied to the cart total.

3.3.10 Use Case: Notifications and Alerts

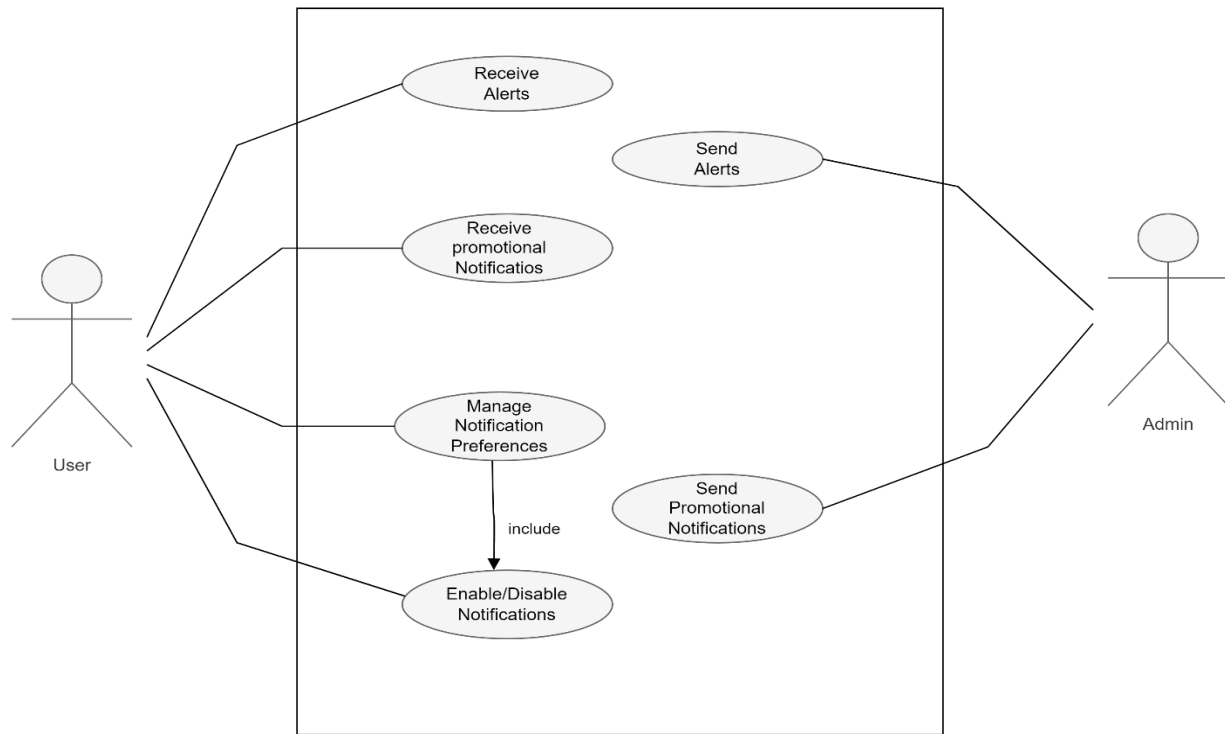


Figure 10 Notifications and Alerts Use Case

Table 10 Notifications and Alerts

Actor	Admin/ User
Pre-condition	<i>User is registered and has opted to receive notifications.</i> <i>User is logged in.</i>
Post-condition	<i>Notification is sent to the user.</i> <i>User's notification preferences are updated.</i>
Steps	1. System triggers a notification event (e.g., order status update). 2. Notification is sent via email, SMS, or in-app.

3.3.11 Use Case: Analytics and Reporting

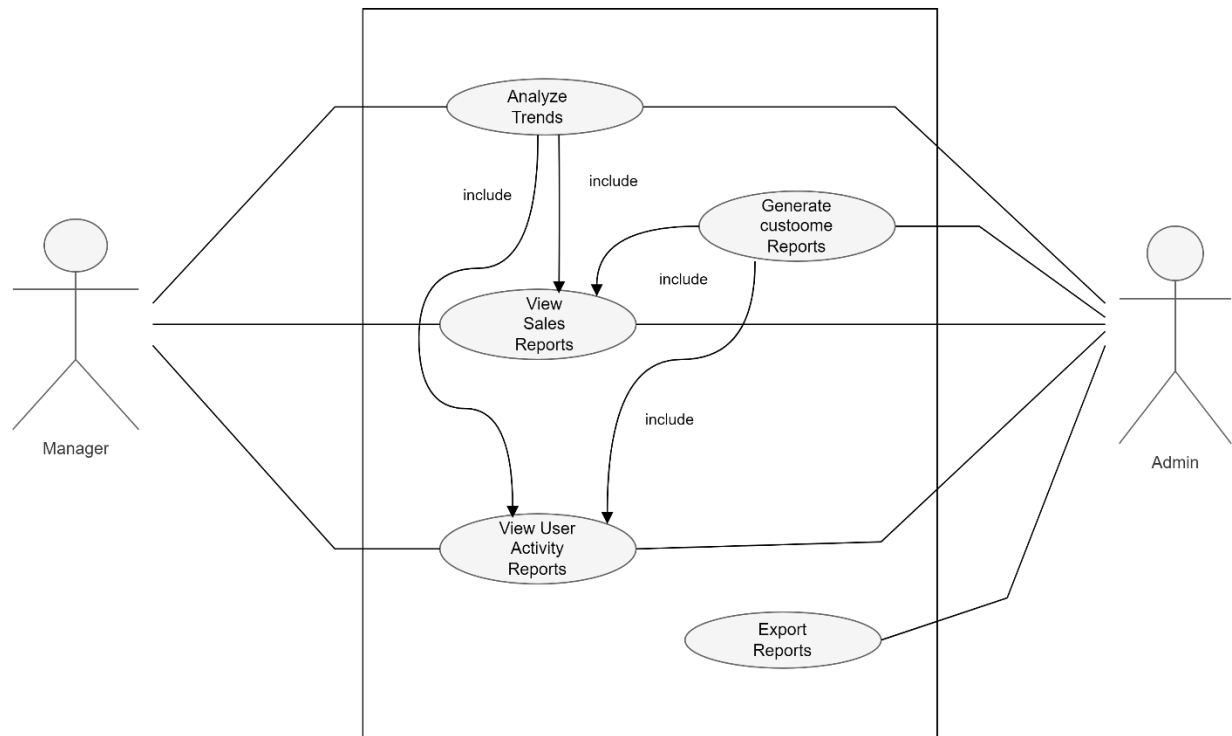


Figure 11 Analytics and Reporting Use Case

Table 11 Analytics and Reporting

Actor	<i>Admin/Manager</i>
Pre-condition	<i>Admin is logged in and has required permissions. Relevant report is generated.</i>
Post-condition	<i>Analytics reports are displayed to the admin. Report is exported in the chosen format (e.g., PDF, CSV).</i>
Steps	<i>1. Admin navigates to the "Analytics" section.</i> <i>2. Selects desired report type (e.g., sales, users).</i> <i>3. System retrieves and displays the report.</i> <i>4. System exports and downloads the report.</i>

3.3.12 Use Case: Security Features

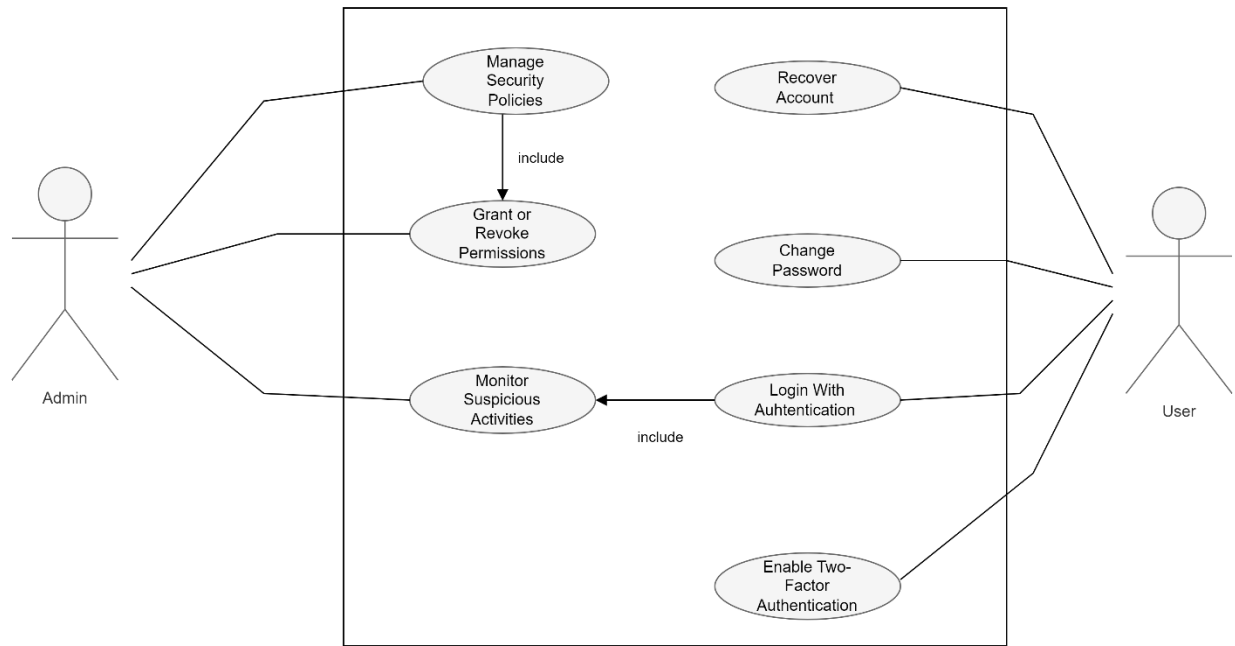


Figure 12 Security Features Use Case

Table 12 Security Features

Actor	<i>User/Admin</i>
Pre-condition	<i>User/Admin is logged in and 2FA is enabled.</i>
Post-condition	<i>Login is verified with 2FA.</i>
Steps	<i>1. User/Admin enters login credentials.</i> <i>2. System prompts for 2FA code.</i> <i>3. User/Admin provides the code.</i> <i>4. System verifies and grants access.</i>

3.3.13 Use Case: Multi-language and Localization Support

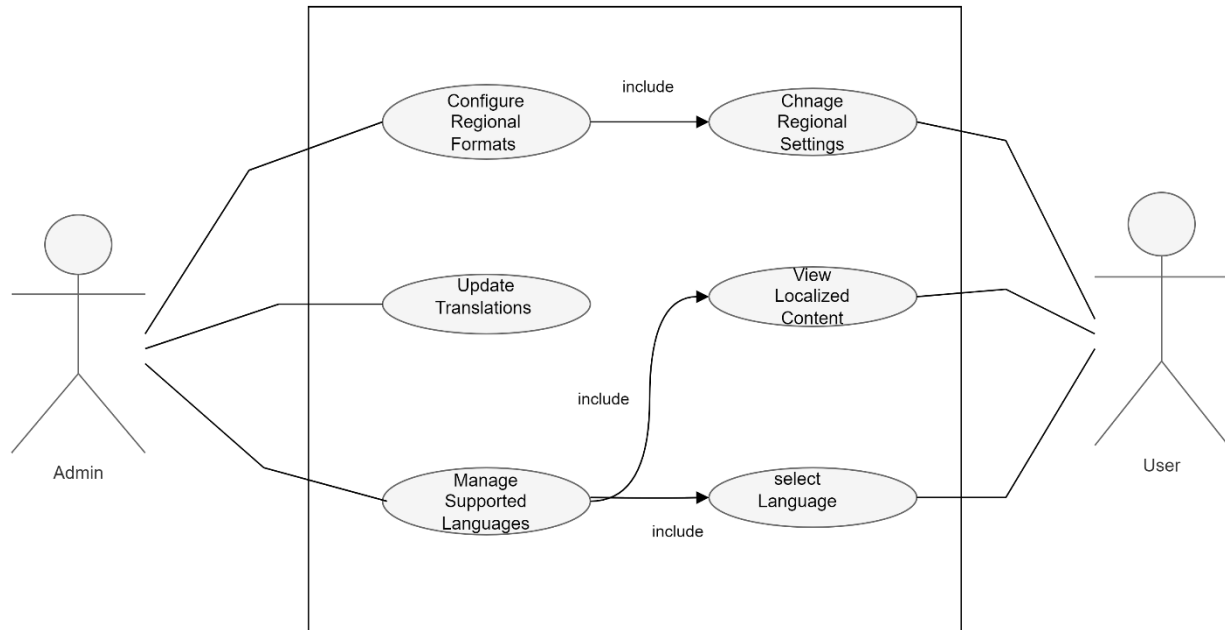


Figure 13 Multi-language and Localization Support Use Case

Table 13 Multi-language and Localization Support

Actor	<i>User</i>
Pre-Condition	<i>User is logged in or browsing the site.</i> <i>User has set a language preference.</i>
Post-Condition	<i>Language preference is updated.</i> <i>Content is displayed in the selected language.</i>
Steps	1. User navigates to the "Language Settings". 2. User selects preferred language. 3. System updates the language across the platform.

3.3.14 Use Case: Feedback and Support System

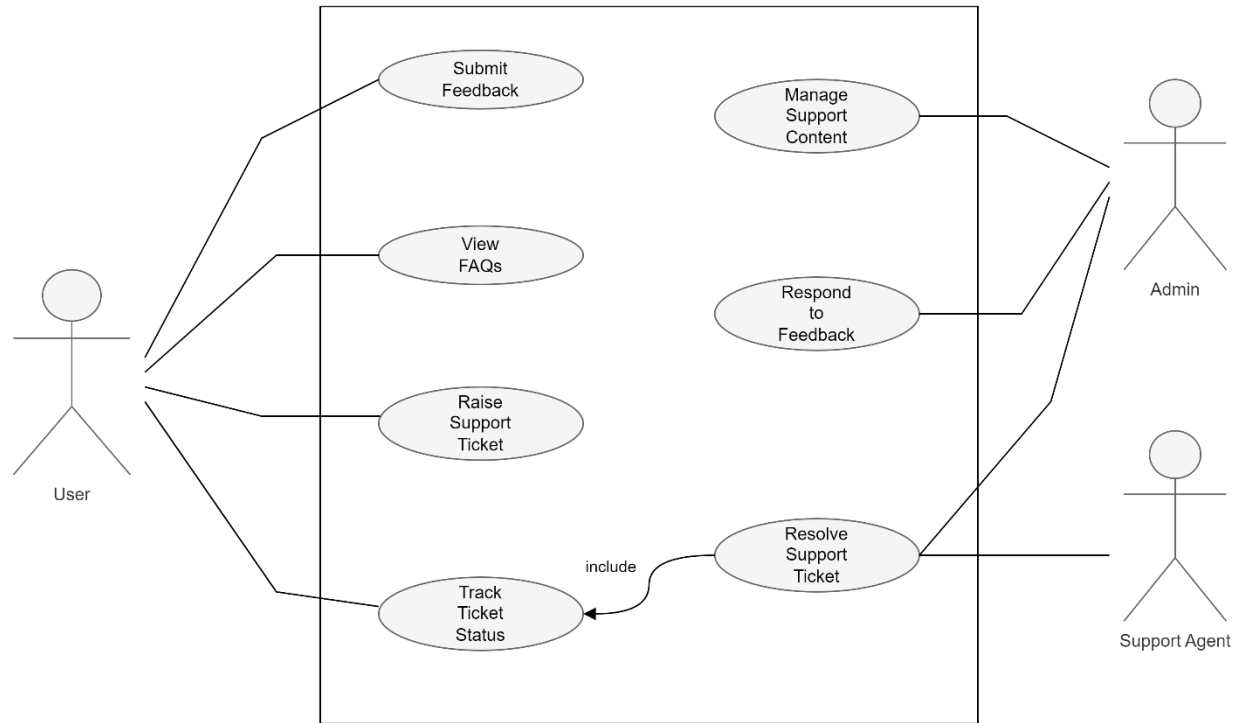


Figure 14 Feedback and Support System Use Case

Table 14 Feedback and Support System

Actor	<i>User/Support Agent/Admin</i>
Pre-Condition	<i>User is logged in or browsing the site. User has set a language preference.</i>
Post-Condition	<i>Language preference is updated. Content is displayed in the selected language.</i>
Steps	<i>1. User navigates to the "Language Settings". 2. User selects preferred language. 3. System updates the language across the platform.</i>

3.3.15 Use Case: Return and Refund Management

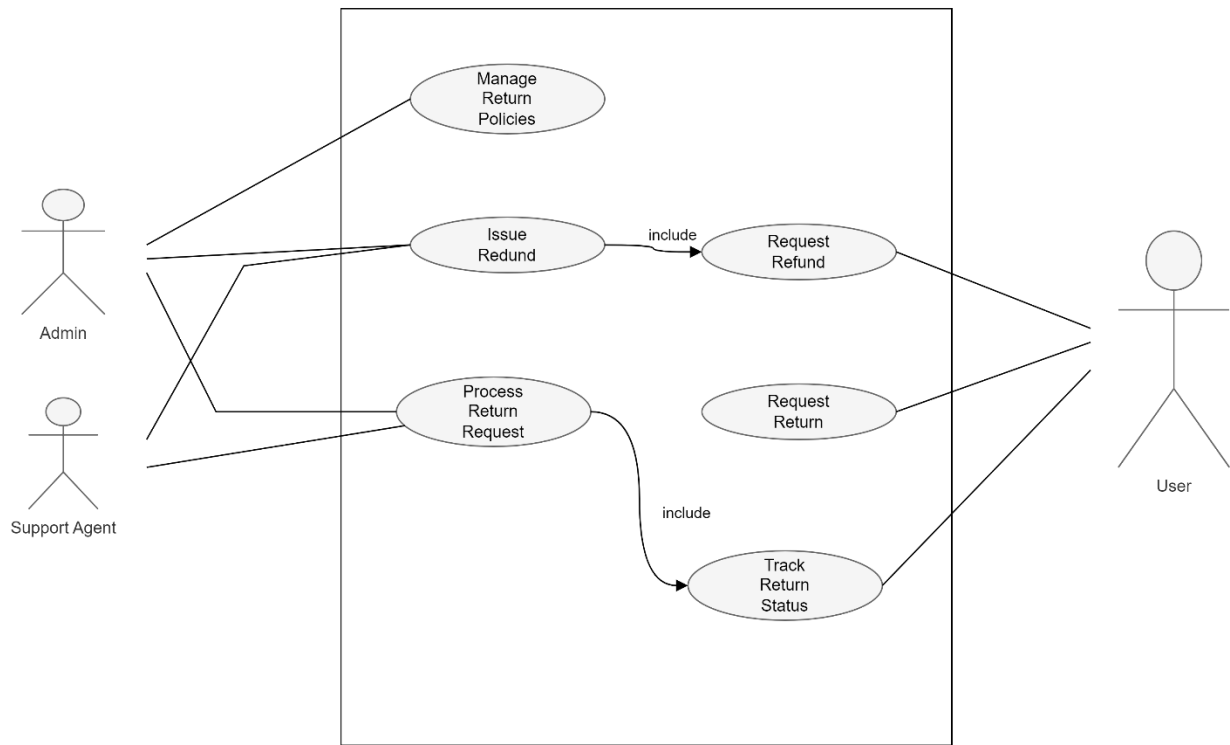


Figure 15 Return and Refund Management Use Case

Table 15 Return and Refund Management

Actor	<i>Admin/Support Agent/User</i>
Pre-Condition	<i>User is logged in or browsing the website. User is logged in or browsing the website.</i>
Post-Condition	<i>Feedback is submitted and recorded. Support request is submitted and ticket is created.</i>
Steps	<i>1. User navigates to "Feedback" section. 2. User enters feedback details. 3. System submits and stores feedback.</i>

3.3.16 Use Case: Social Shopping with Live Interaction

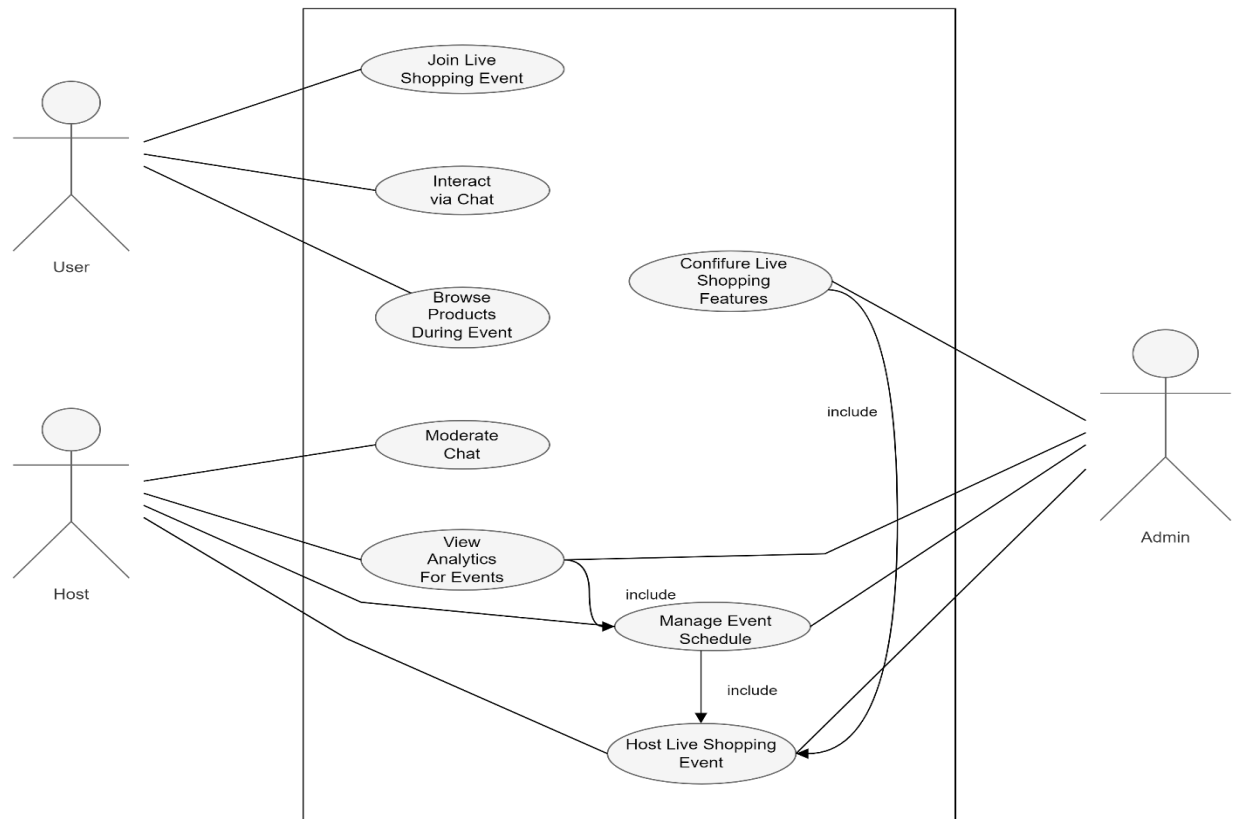


Figure 16 Social Shopping with Live Interaction Use Case

Table 16 Social Shopping with Live Interaction

Actor	<i>Admin/User/Host</i>
Pre-Condition	<i>User is logged in and has access to live shopping sessions.</i>
Post-Condition	<i>User is participating in a live shopping session.</i>
Steps	<i>1. User navigates to the live shopping section.</i> <i>2. User selects a live session.</i> <i>3. User joins and watches the session.</i>

3.3.17 Use Case: Product Review and Ratings

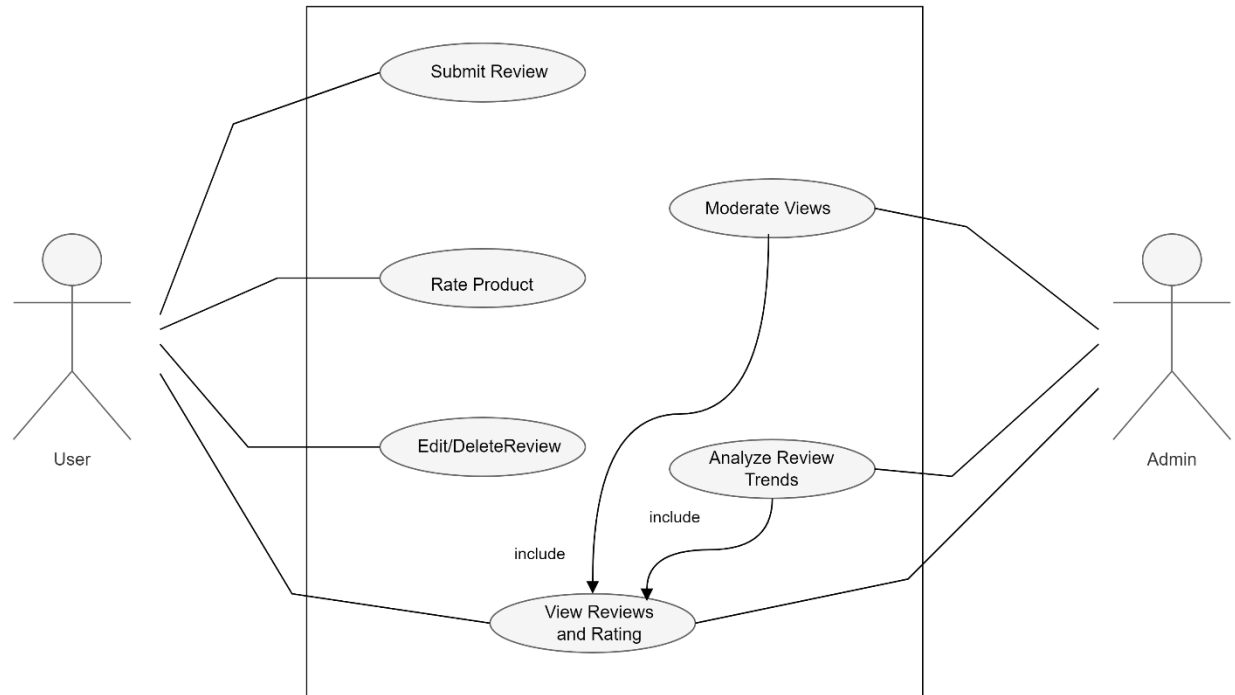


Figure 17 Product Reviews and Ratings Use Case

Table 17 Product Reviews and Ratings

Actor	<i>User/Admin</i>
Pre-Condition	<i>User has purchased a product or is logged in. Product has existing reviews and ratings.</i>
Post-Condition	<i>Review and rating are submitted and recorded. Reviews and ratings are displayed to the user.</i>
Steps	<i>1. User navigates to the product page. 2. User selects the "Write a Review" option. 3. User provides a rating and review. 4. System saves the review and displays it under the product.</i>

3.3.18 Use Case: Customer Support and FAQs

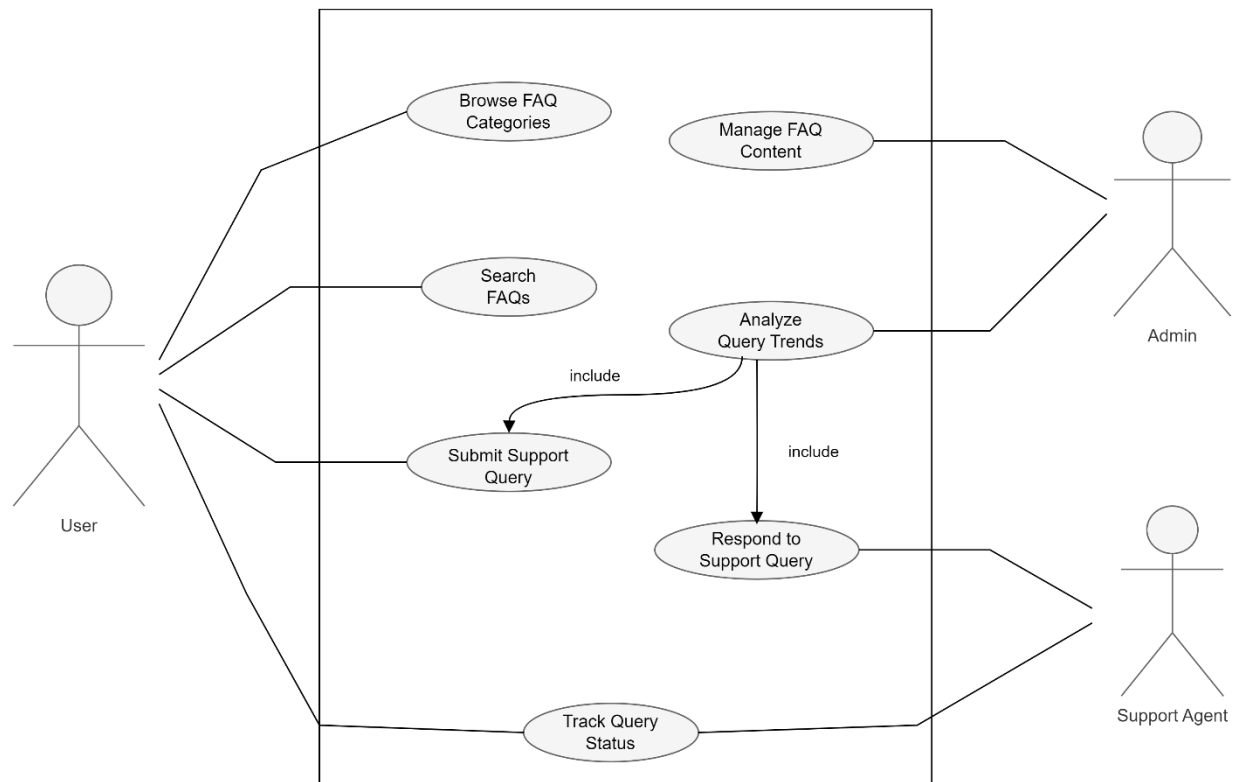


Figure 18 Customer Support and FAQs Use Case

Table 18 Customer Support and FAQs

Actor	User/Admin/Support Agent
Pre-Condition	User is on the website or mobile app. A support request has been submitted by the user.
Post-Condition	FAQs are displayed to the user. User receives a response to their query.
Steps	1. User navigates to the "Help" or "FAQs" section. 2. System retrieves and displays categorized FAQs.

3.3.19 Use Case: Multi-Destination Checkout



Figure 19 Multi-Destination Checkout Use case

Table 19 Multi-Destination Checkout

Actor	<i>User/Admin</i>
Pre-Condition	<i>User has products in the shopping cart.</i>
Post-Condition	<i>Order is placed with items shipped to multiple destinations.</i>
Steps	<i>1. User navigates to the checkout page.</i> <i>2. User selects "Ship to multiple destinations" option.</i> <i>3. User assigns shipping addresses to specific items.</i> <i>4. User reviews the order summary.</i> <i>5. User confirms payment and places the order.</i>

3.4 Classes / Objects

This section contains major classes of the GIFT GALLERY(Gift and Accessories Store) system.

3.4.1 *Login*

Login
-emailId: string -password: string
+setEmailId(email: string) +getEmailId(): string +setpassword(psw: string) +getpassword(): string

3.4.2 *Registration*

Registration
-Name: string -fatherName: string -email: string -phoneNo: int -password: string -gender: string
+getName() +setName() +getpassword() +setpassword() +create-newAccount()

3.4.3 *Feedback*

Feedback
-feedbackId: int -userId: int -message: string -rating: int
+setFeedbackId(id: int): void +getFeedbackId(): int +setUserId(id: int): void +getUserId(): int +setMessage(msg: string): void +getMessage(): string +setRating(rating: int): void +getRating(): int

3.4.4 *Database*

Database
+dbname +dbURL +userName +password
+InsertRecord() +updateRecord() +deleteRecord() +fetchRecord() +createBackup() +restoreFromBackup()

3.4.5 *Order*

Order
-orderId-int -product-int -userId-int
+addOrder() +deleteOrder() +placeOrder() +getOrder() +setOrder()

3.4.6 *Shopping cart*

ShoppingCart
+productId: int +shippingId: int +quantity: int +cost: int
+addToCart() +deletefromCart() +viewCart()

3.4.7 *Product*

Product
+serialNumber +productName: string +password: int +productPrice: int
+addProduct() +deleteProduct() +updateProduct() +searchProduct() +buy() +display()

3.4.8 *Admin*

Admin
+name: string +emailId: string +password: int
+registration() +login() +addRecord() +deleteRecord() +updateRecord() +resetPassword() +cashOnDelivery() +costCalculation() +logout()

3.4.9 Customer

Customer
+name: string +emailId: string +password; int +address: string +creditInfo: int
+registration() +login() +viewCart() +addProduct() +deleteProduct() +order() +resetPassword() +logout()

3.4.10 User

User
-firstName -lastName -dob -gender -email +Attribute 1
+viewPersonalInformation() +updateInfo() +deleteInfo()

3.5 Non-Functional Requirements

3.5.1 Performance

The website must load within **2 seconds** for 95% of users under normal traffic conditions.

3.5.2 Reliability

The website should maintain **99.9% uptime** on a monthly basis.

3.5.3 Availability

The platform should be operational **24/7**, with planned maintenance causing no more than **1 hour of downtime per month**.

3.5.4 Security

All sensitive data, such as payment information, must be encrypted using **AES-256**.

The system should enforce **two-factor authentication (2FA)** for administrator access.

3.5.5 Maintainability

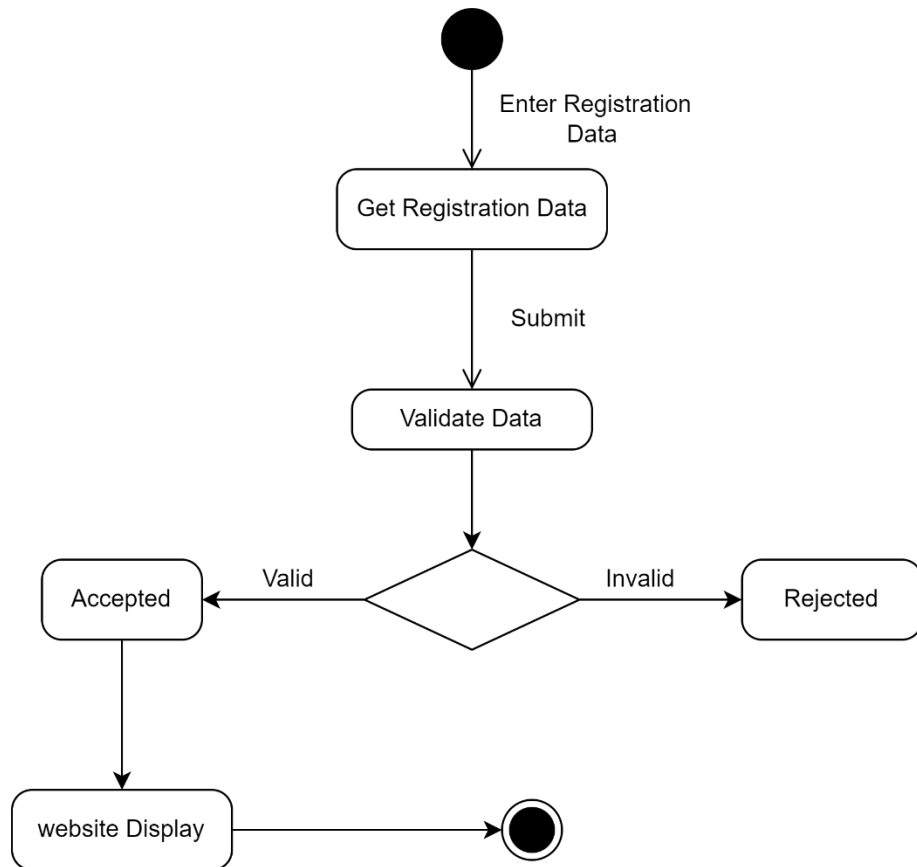
The codebase should follow **modular design principles** to allow new features to be integrated within **1 week**.

3.5.6 Portability

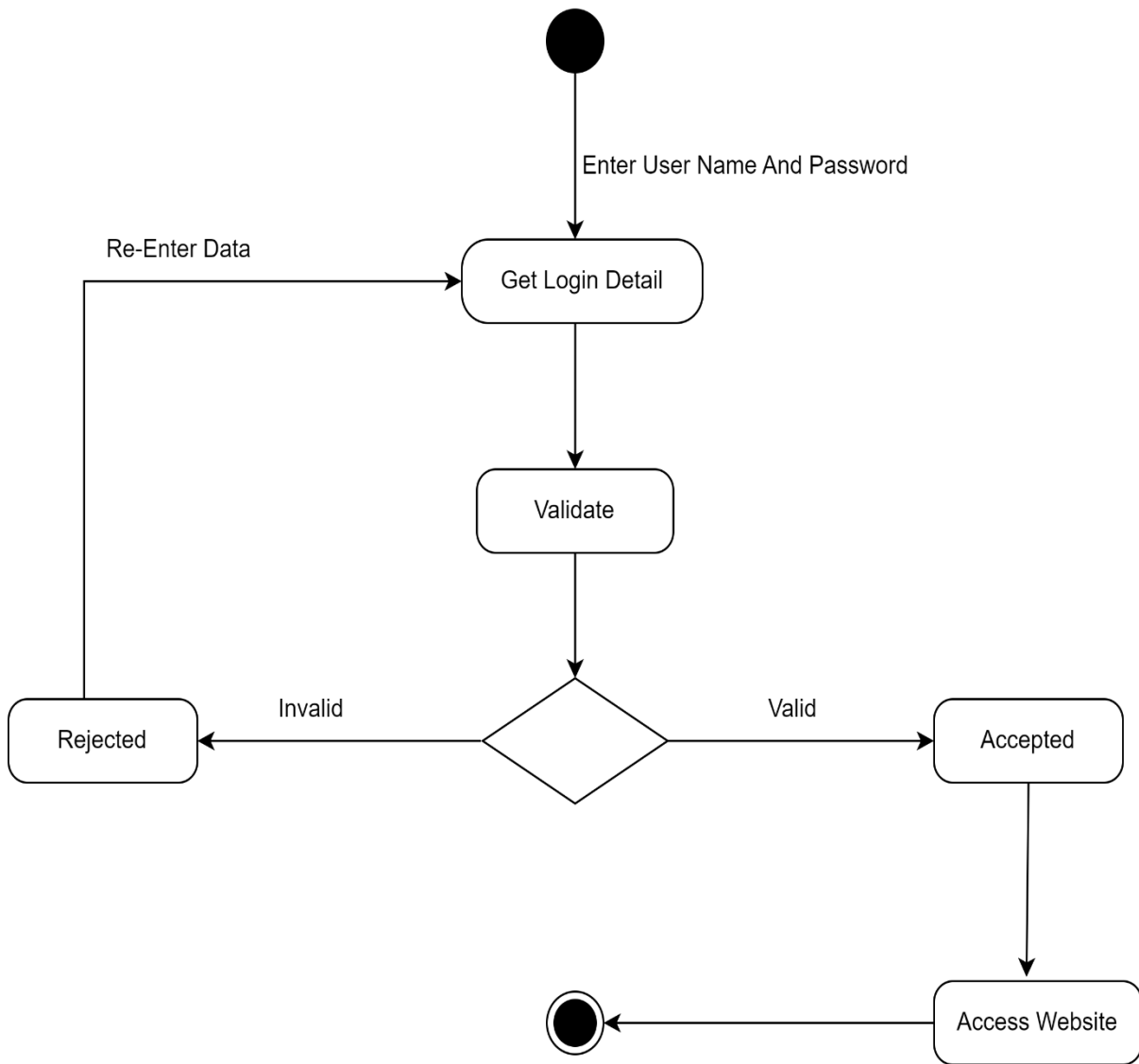
The website should be compatible with major browsers (Chrome, Firefox, Safari, Edge) and mobile devices.

3.6 Activity Diagram

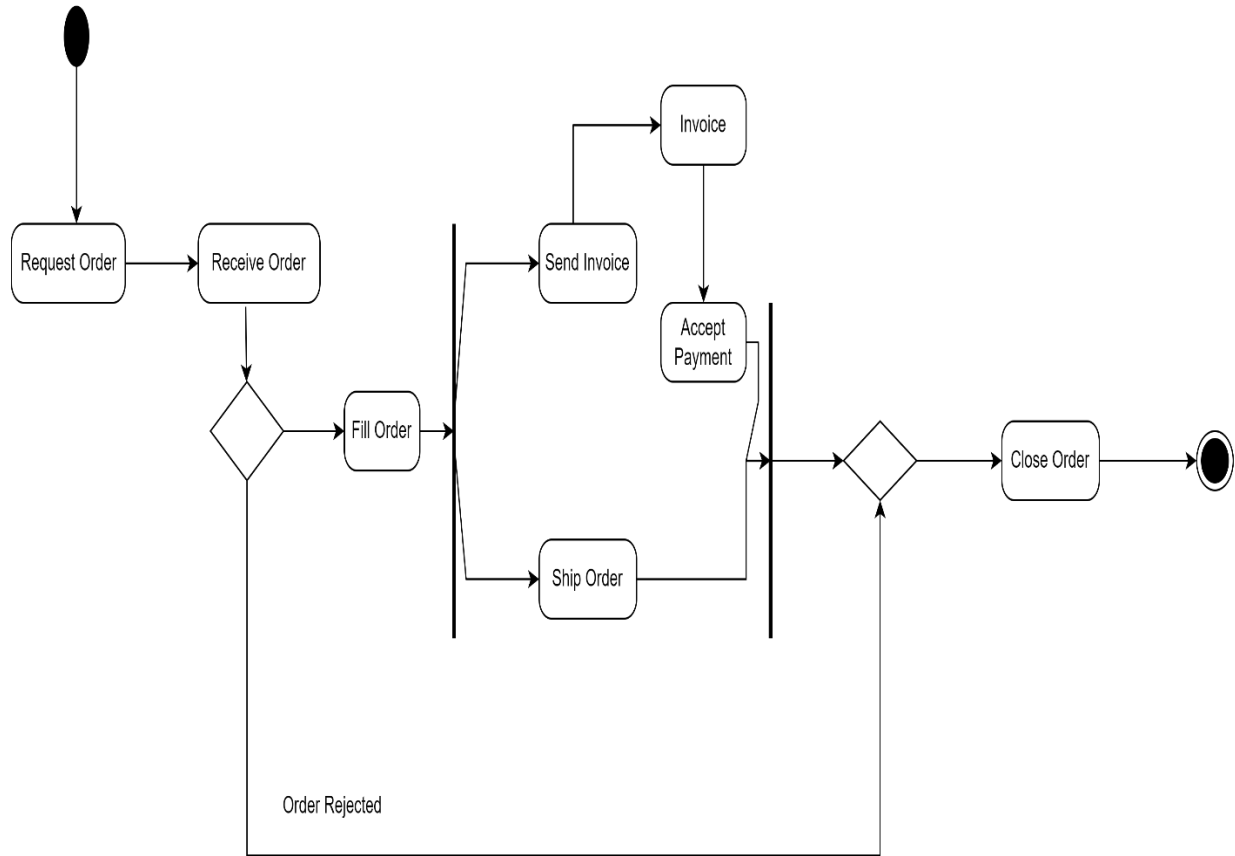
3.6.1 Activity Diagram of Registration



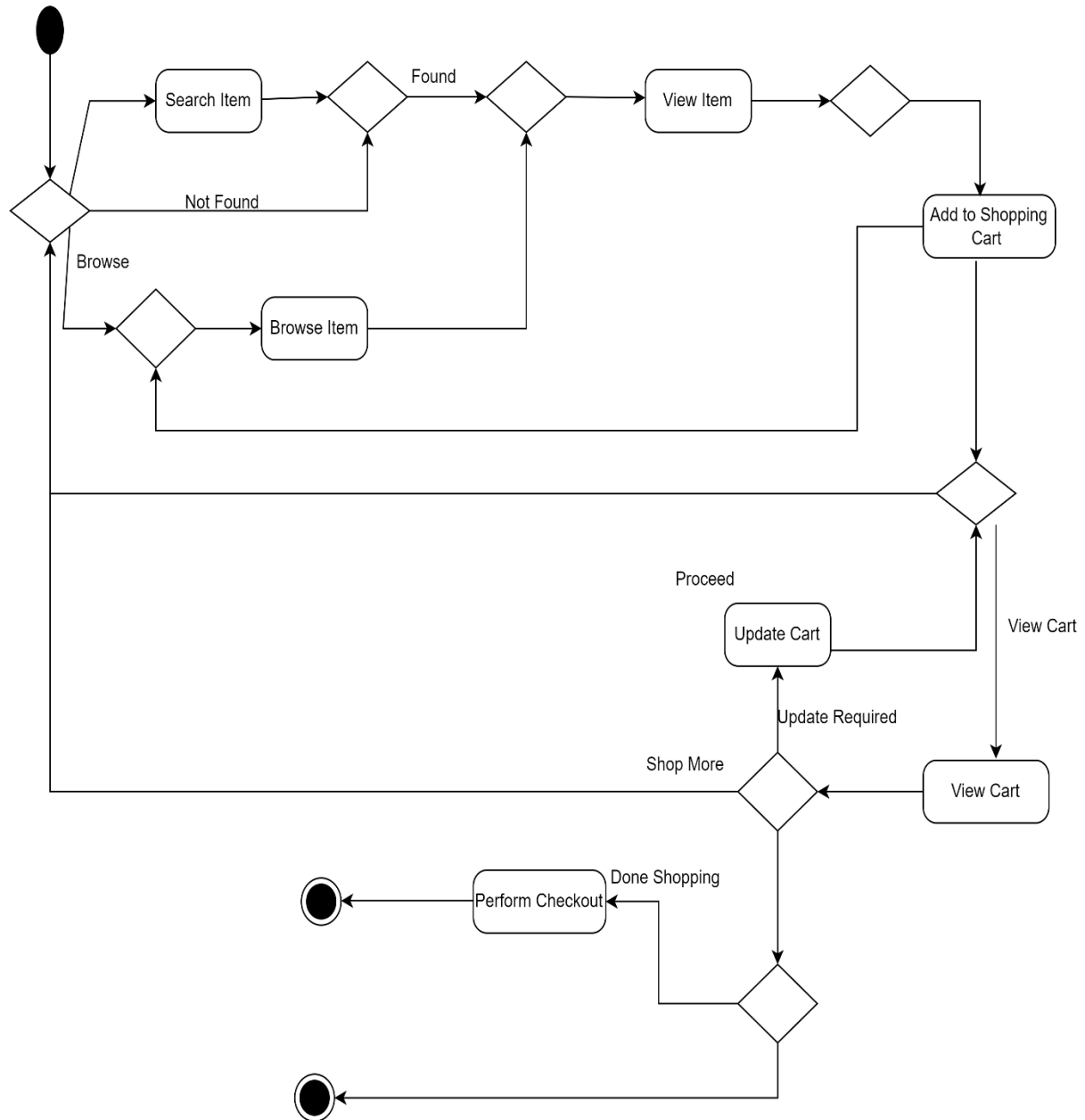
3.6.2 Activity Diagram of Login



3.6.3 Activity Diagram of Order Process

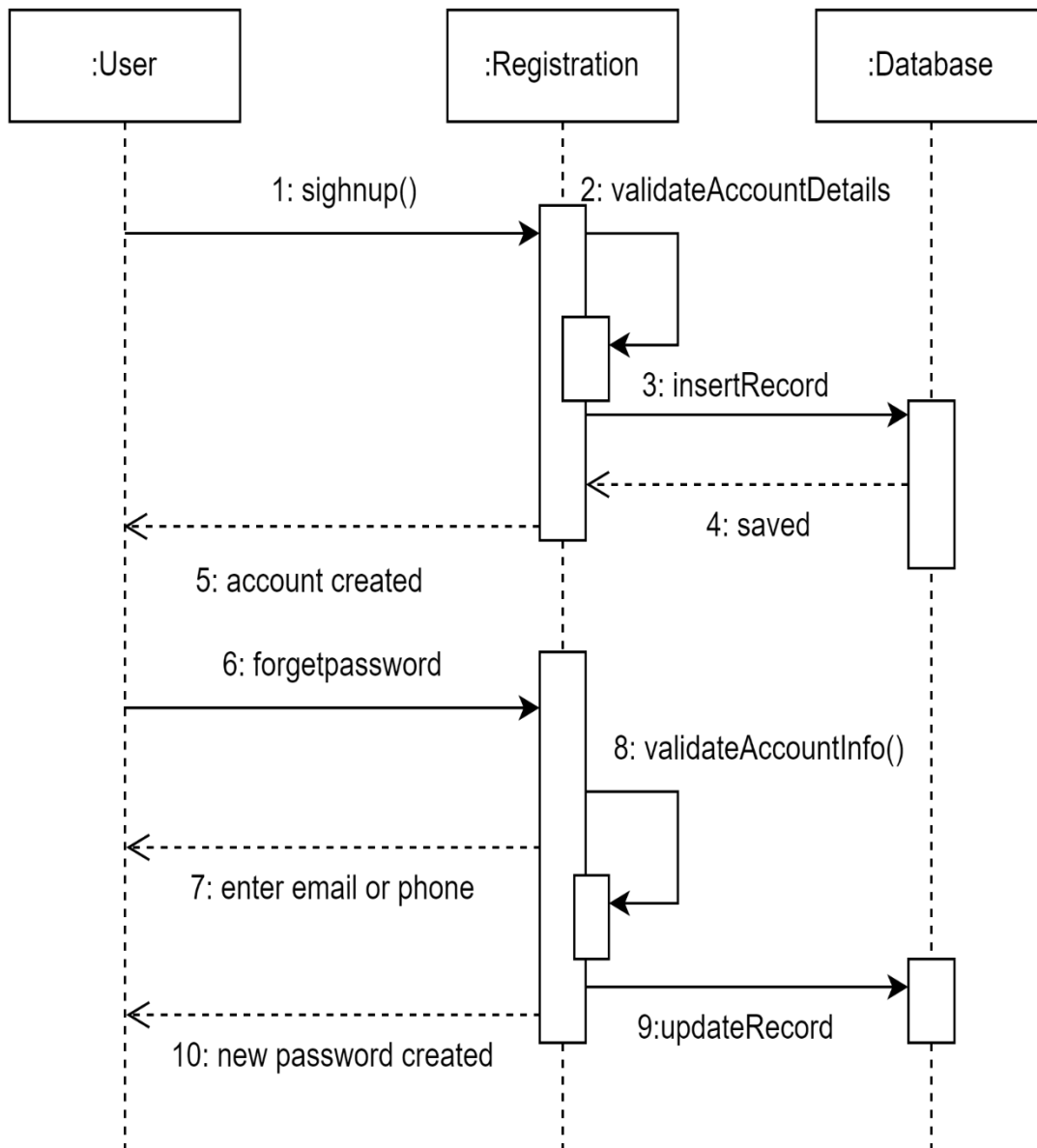


3.6.4 Activity Diagram of Shopping Cart

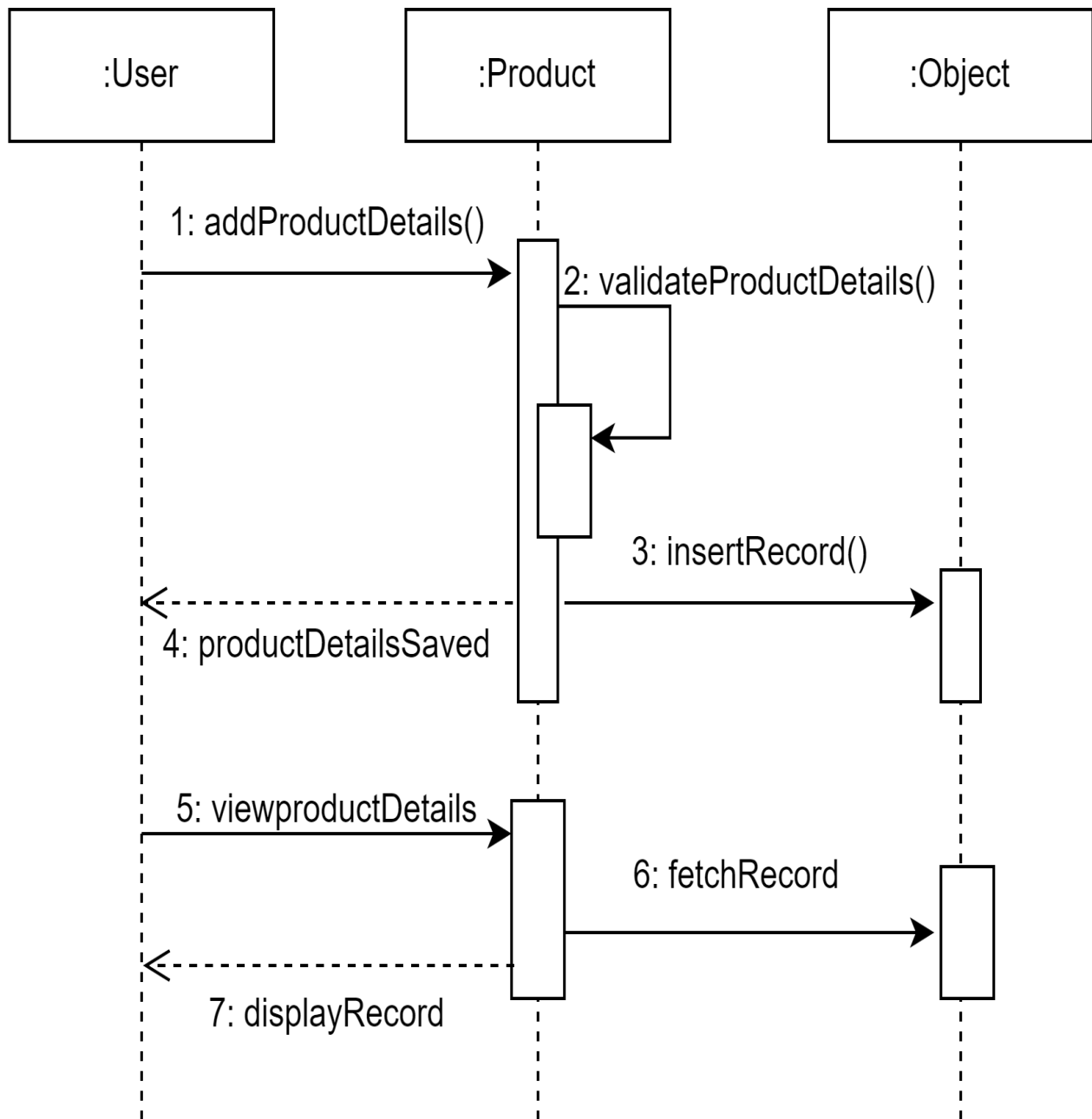


4.1 Sequence Diagrams

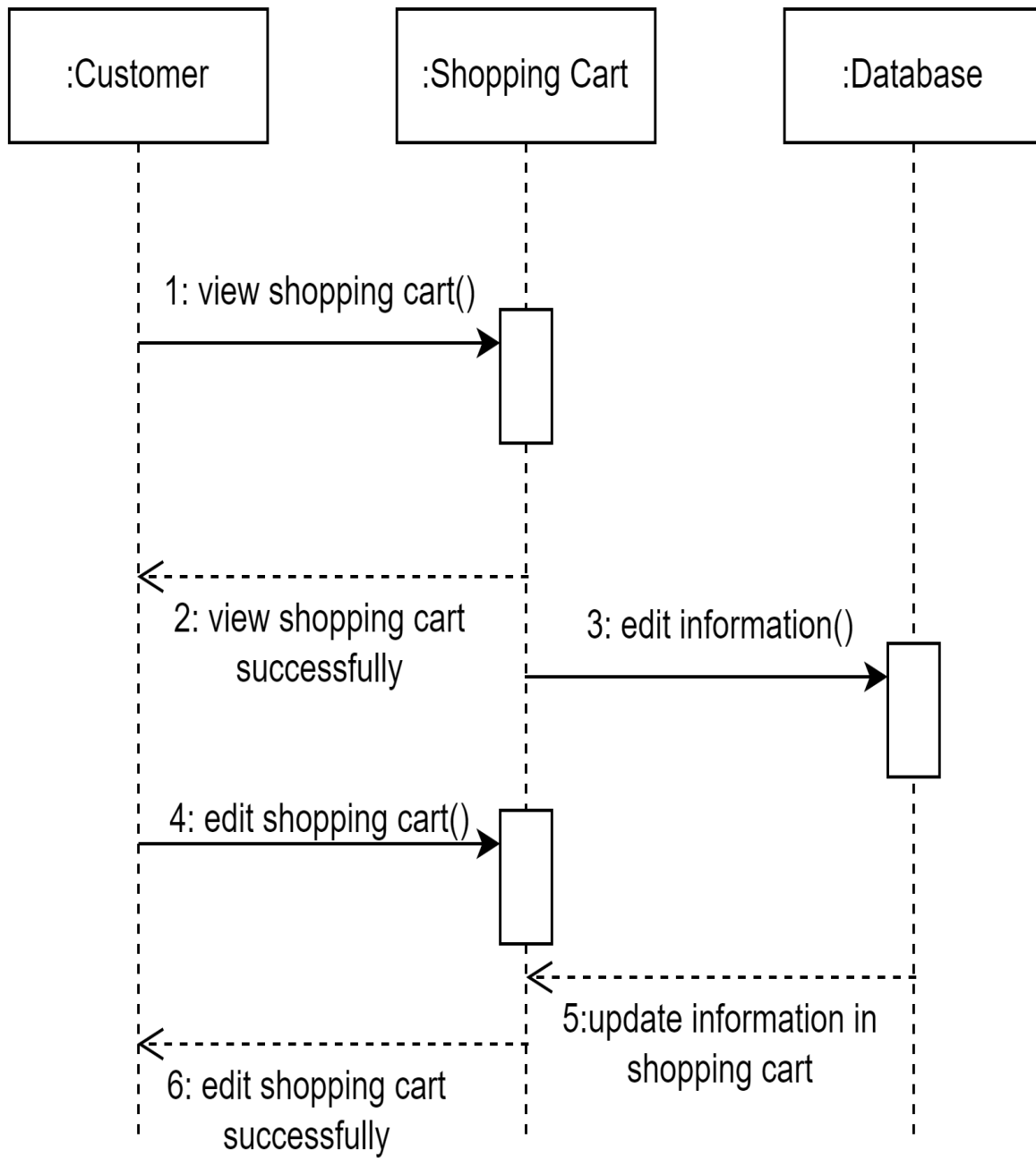
4.1.1 Sequence Diagrams of Registration

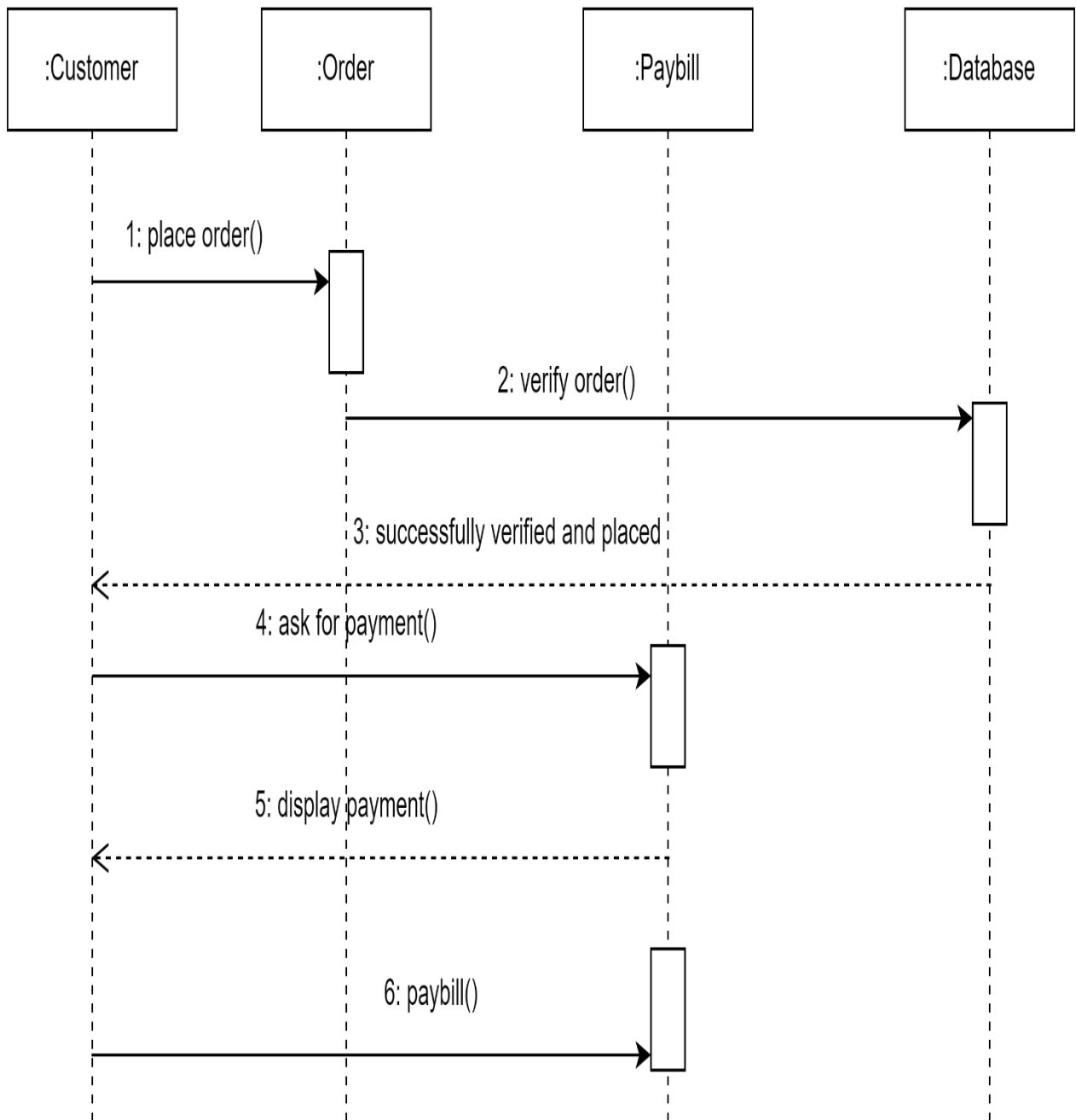


4.1.2 Sequence Diagrams of Add Product



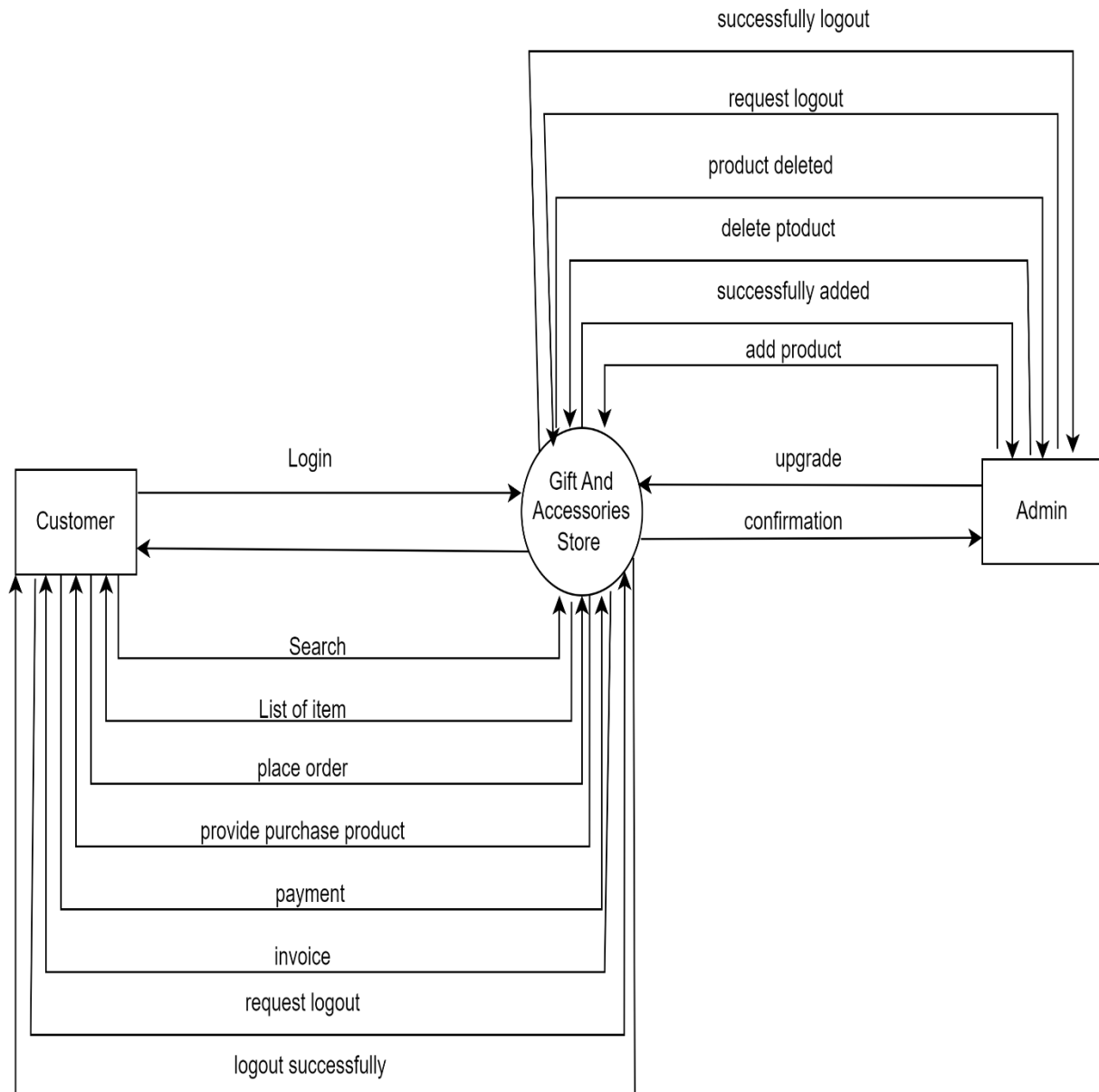
4.1.3 Sequence Diagrams of Shopping Cart



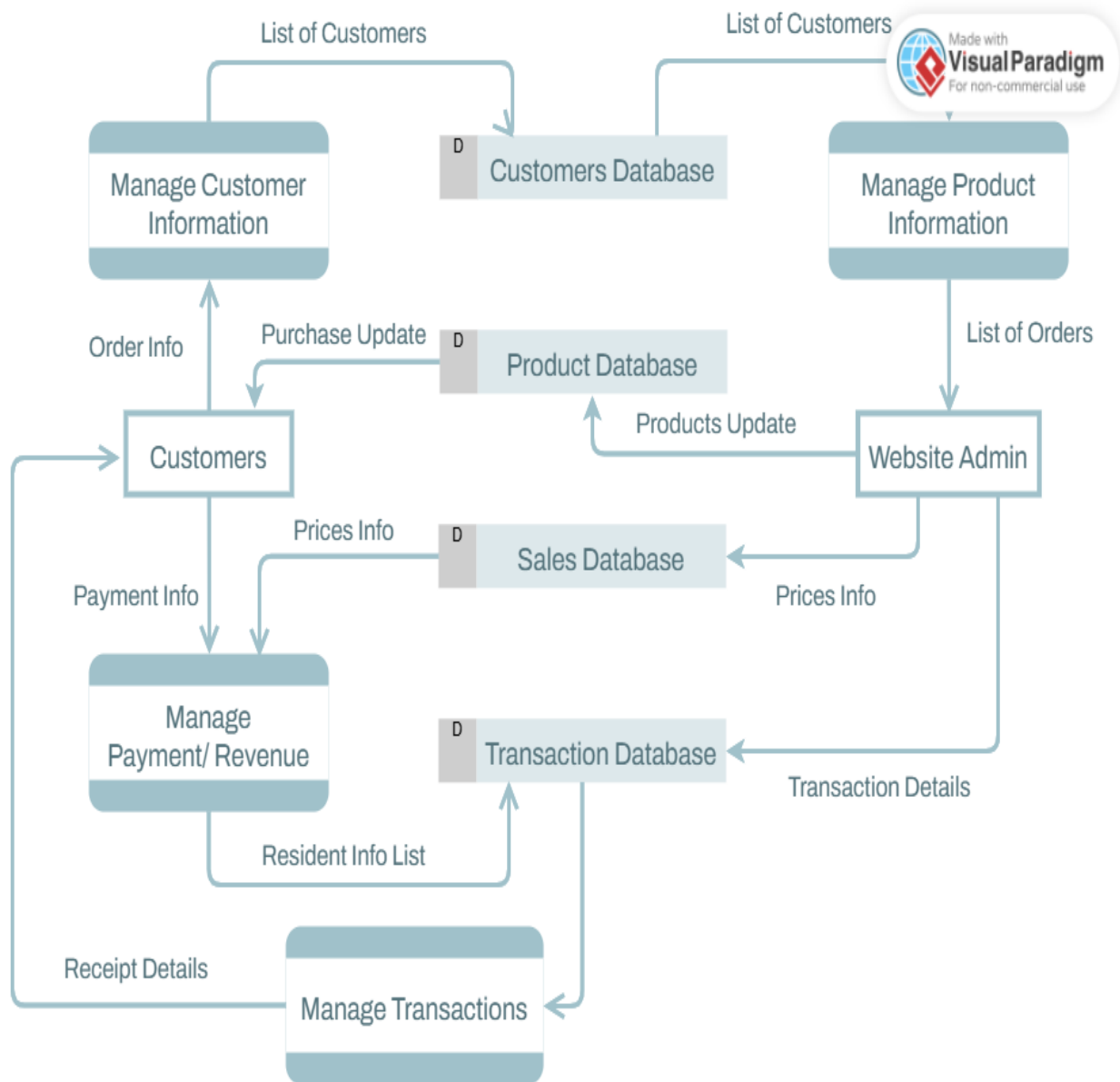
4.1.4 Sequence Diagrams of Payment

4.2 Data Flow Diagrams (DFD)

4.2.1 Level 0 Data Flow Diagrams (DFD)

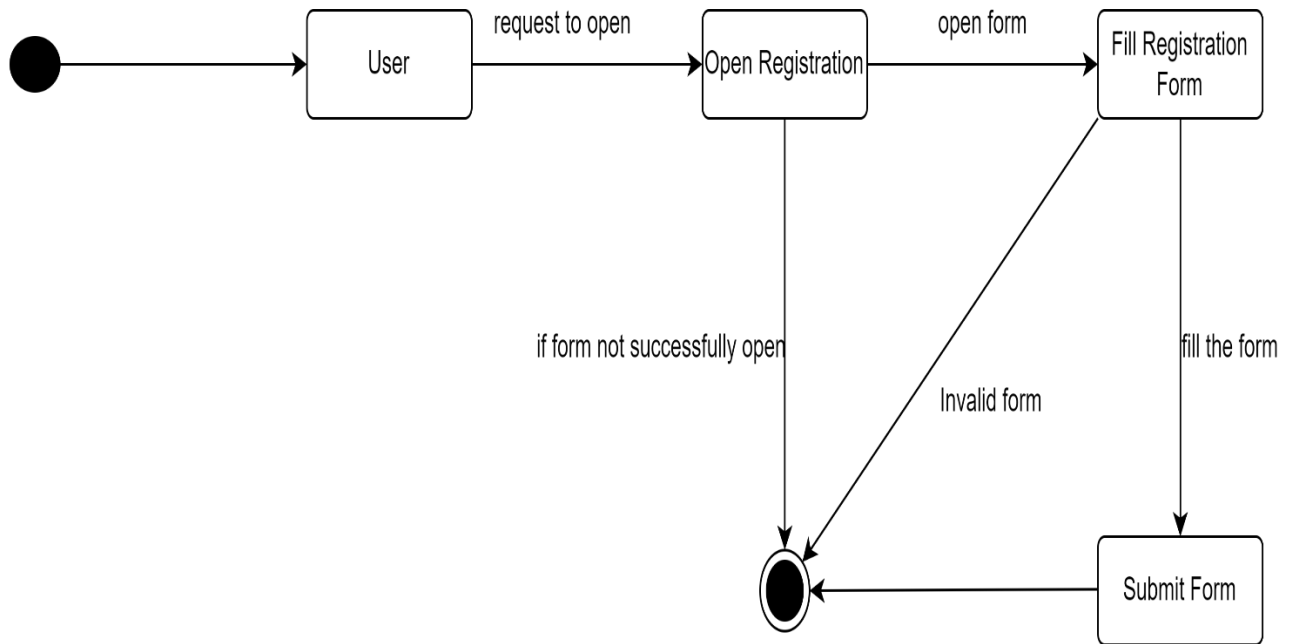


4.2.2 Level 1 Data Flow Diagrams (DFD)

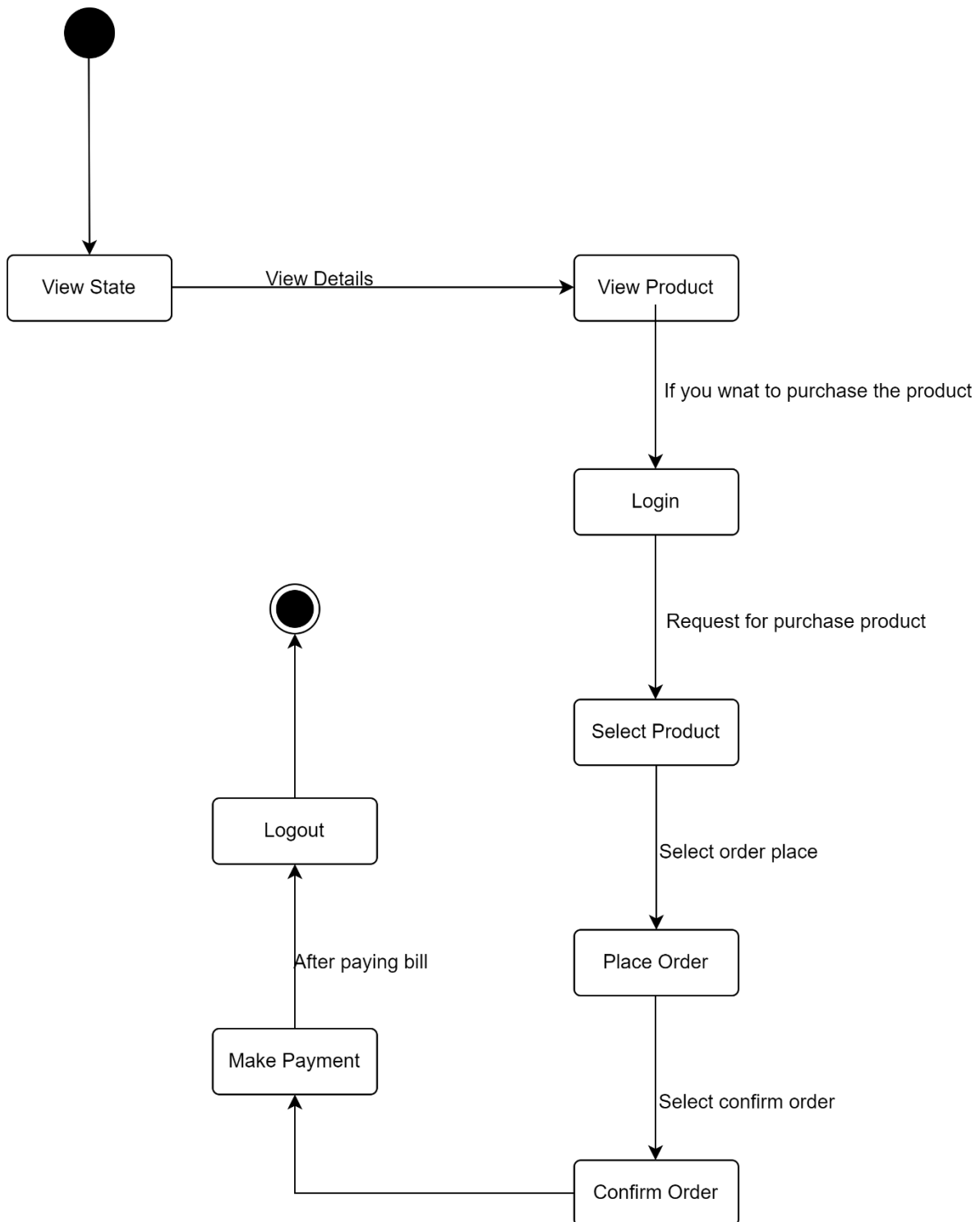


4.3 State-Transition Diagrams (STD)

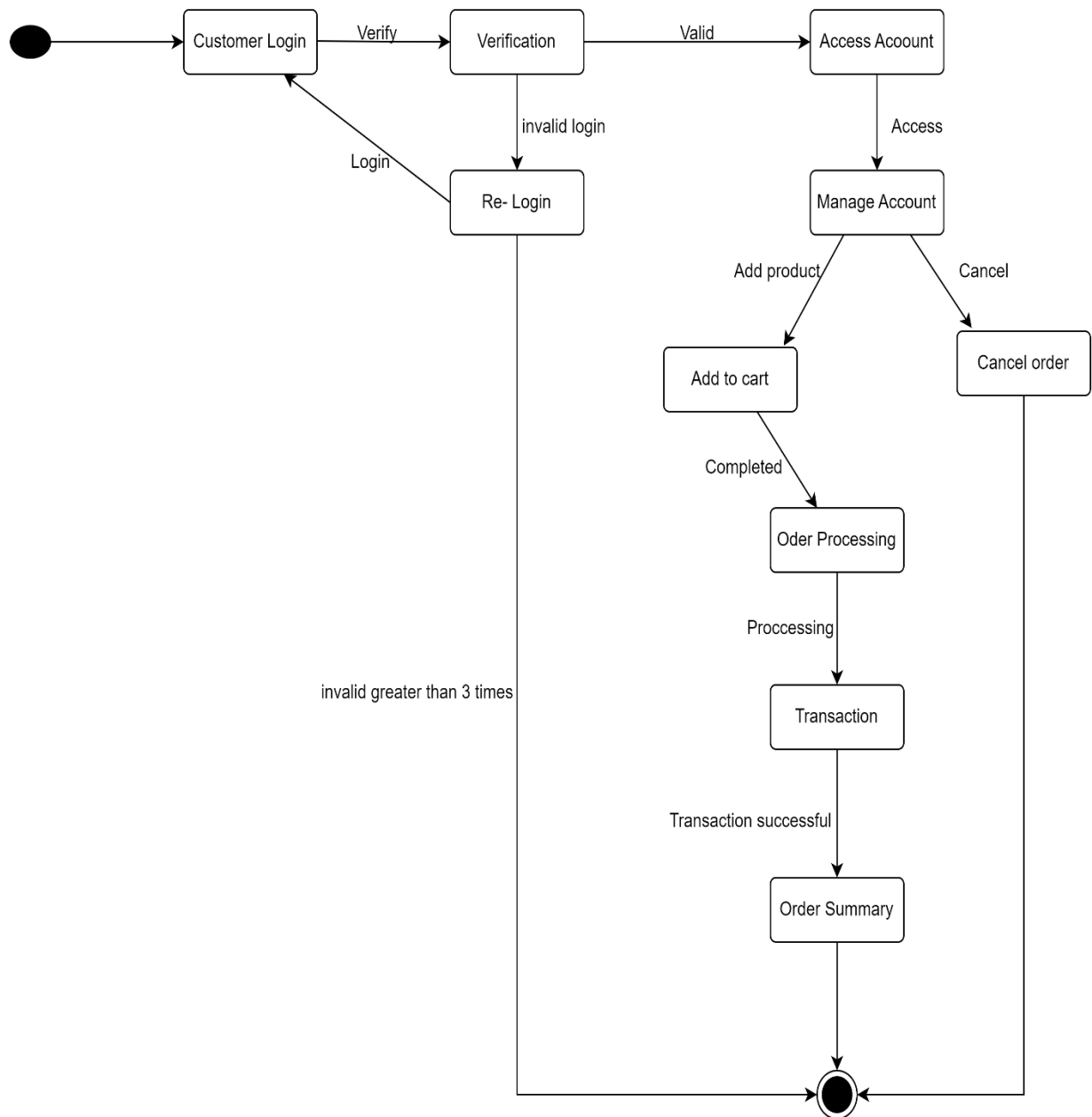
4.3.1 State-Transition Diagrams of Registration



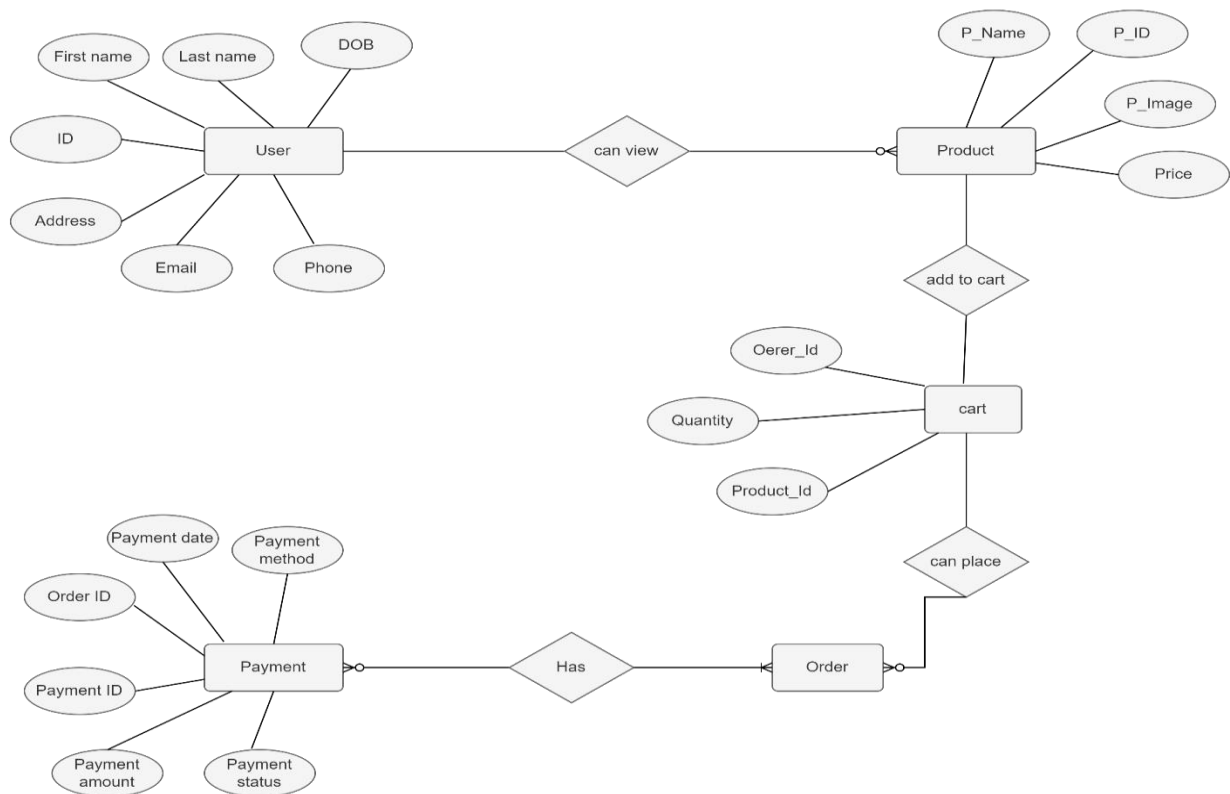
4.3.2 State-Transition Diagrams of Product



4.3.3 State-Transition Diagrams of Customer



4.4 ER Diagram



5. Supporting Information

Appendix A – Background Research on:

- Topic 1
- Topic 2
- Topic 3
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- Topic *n*

Appendix B – Data Dictionary